Bulletin of Earthquake Engineering
https://doi.org/10.1007/s10518-026-02639-x

ORIGINAL ARTICLE

Optimal intensity measures for fragility assessment of bridge components under mainshock–aftershock seismic sequences in subduction zones

Brian Cagua[1] · Gerard J. O'Reilly[2] · Jawad Fayaz[3] · Rodrigo Astroza[1]

Received: 21 January 2026 / Accepted: 9 August 2026
© The Author(s), under exclusive licence to Springer Nature B.V. 2026

Abstract
Bridges in seismically active regions are exposed to mainshock–aftershock (MS–AS) sequences, yet most fragility models either neglect AS effects or rely on intensity measures (IMs) developed for single events, resulting in overly optimistic predictions. This study introduces a framework that identifies optimal IMs for bridge components under MS–AS sequences, addressing a gap largely untested in subduction environments. The methodology combines seismic disaggregation to define site-specific scenarios with a stochastic AS model tailored to subduction zones, generating 10,000 physically consistent MS–AS sequences for each MS scenario. Conditional IM vectors are generated using a recurrent neural network-based ground motion model, calibrated for the Chilean subduction context, which preserves observed MS–AS spectral correlation. Ground motions are selected through multi-objective optimization to ensure hazard consistency. The framework is demonstrated on the Águila Norte Bridge in Chile through 12,000 nonlinear dynamic analyses capturing structural response under MS–AS excitation. Engineering demand parameters for elastomeric bearings, abutments, and columns are defined using both traditional indices and residual-capacity-based measures. Eleven IMs are evaluated using four complementary metrics—efficiency, practicality, proficiency, and sufficiency. Results consistently rank Housner Intensity (HI), average spectral acceleration ($Sa_{avg}(T_1)$), and orientation-independent spectral acceleration ($Sa_{RotD50}(T_1)$) ahead of peak ground acceleration (PGA) and spectral acceleration at the fundamental period ($Sa(T_1)$), with R^2 improving by 8–14% and selected optimal IMs reducing dispersion β by more than 50% relative to PGA for the elastomeric-bearing response. Logistic exceedance probability surfaces further confirm that neglecting ASs systematically underestimates component-level damage exceedance probabilities.

Keywords Mainshock-aftershock sequence · Optimal intensity measure · Seismic fragility · Bridges · Subduction zones

Extended author information available on the last page of the article

Published online: 22 August 2026

1 Introduction

Bridges are critical components of transportation networks and their uninterrupted functionality is essential for post-earthquake response, economic stability, and community recovery. In subduction zones bridges are often subjected to complex demands from mainshock–aftershock (MS–AS) sequences. Such loading can accelerate deterioration, reduce residual capacity, and, in severe cases, cause collapse. Field evidence from the 2010 Maule-Chile, 2011 Tohoku-Japan, and 2023 Kahramanmaraş-Turkey earthquakes documents failures including span unseating, column shear and flexural damage, bearing failure, and approach fill settlement (Kawashima et al. 2011; Kawashima and Buckle 2013; Shirato and Odawara 2024). In several cases, ASs were identified as aggravating or triggering factors of collapse, particularly where MS-induced damage had reduced stiffness and energy dissipation capacity (Ruiz-García and Negrete-Manriquez 2011; Li Yue et al. 2014; Guo et al. 2020). These observations highlight the need to explicitly consider AS effects in probabilistic fragility assessment of bridge infrastructures.

MS–AS sequences in subduction zones exhibit distinctive temporal, spatial, and spectral patterns governed by tectonic regime, focal depth, and rupture properties (Davis and Frohlich 1991; Wiens and McGuire 1995; Kagan 2002; Zhang et al. 2020). Because AS distributions often scale with MS rupture dimensions, their cumulative effects become particularly critical in subduction-zone settings, where long-duration sequences and multiple large ASs are common. Despite this relevance, design codes and fragility practices rarely incorporate MS–AS explicitly, often assuming independent single-event ground motions (GMs). Conventional intensity measures (IMs)—such as peak ground acceleration (PGA) or spectral acceleration at the fundamental period, $Sa(T_1)$—capture only the response to a single excitation and therefore fail to represent the sequential, cumulative, and phase-dependent demands imposed by successive MS–AS events. This limitation can lead to non-conservative fragility estimates (O'Reilly 2021a, b; Yu et al. 2021). Overcoming this limitation requires probabilistic frameworks that integrate MS–AS sequence characteristics into hazard modelling, conditional GM selection, and IM evaluation to improve vulnerability estimates for critical infrastructures such as bridges in subduction zones.

From a probabilistic standpoint, the explicit treatment of MS–AS sequences has been progressively incorporated into AS-conditioned probabilistic seismic hazard assessment (APSHA), reliability, and state-dependent damage/risk-assessment frameworks. APSHA characterizes the conditional occurrence and intensity of ASs following a triggering MS, while state-dependent damage/risk assessment propagates the post-MS structural condition into subsequent demand, damage, fragility, and risk evaluation. This distinction has been addressed through AS hazard and adaptive risk formulations (Yeo and Cornell 2005, 2009; Ebrahimian et al. 2014; Jalayer and Ebrahimian 2017), as well as through post-MS risk models, Markov risk-assessment frameworks, state-dependent fragility curves, and cumulative-damage-based risk approaches (Shokrabadi and Burton 2018; Orlacchio et al. 2024; García de Quevedo et al. 2025). More recently, MS–AS fragility methodologies have also been extended to buildings and bridges using probabilistic demand models, data-driven formulations, and community-resilience-oriented fragility models (Zhou et al. 2021; Chen et al. 2024; Harati and van de Lindt 2024).

At the structural-analysis level, one persistent challenge is the accurate modeling of expected MS–AS sequences and the assessment of structures in a post-MS damage state. This

issue has been addressed in early sequential assessments of steel and reinforced-concrete systems (Luco et al. 2004; Lee and Foutch 2004; Fragiacomo et al. 2004; Li and Ellingwood 2007; Ruiz-García 2012). Building on the incremental dynamic analysis concept (Vamvatsikos and Cornell 2002), MS–AS incremental dynamic analyses have been employed to derive fragilities of MS-damaged structures (Luco et al. 2011; Ryu et al. 2011). Subsequent studies have further examined repeated-earthquake response, nonlinear response potential, vulnerability assessment, double incremental dynamic analysis, and component- or system-level MS–AS fragility formulations for buildings and bridges (Hatzigeorgiou and Beskos 2009; Goda 2012; Wen et al. 2017; Di Trapani and Malavisi 2019). Collectively, these studies indicate that MS–AS sequences should be constructed from real recorded sequences or from well-supported, hazard-consistent synthetic generation that preserves the dependence of AS characteristics on the conditioning MS, rather than from arbitrary pairing of independent single-event records. This issue is also reflected in previous discussions on sequential record-selection strategies, where MS–MS, target-directed MS–MS, same-sequence MS–AS, and different-sequence MS–AS approaches provide different levels of consistency in reproducing the magnitude, distance, intensity, and spectral dependence between events. This requirement directly motivates the sequence-generation strategy adopted in this study.

Although MS–AS fragility has been examined for RC frames (Di Trapani and Malavisi 2019; Afsar Dizaj et al. 2022) and lifeline systems such as pipelines and transmission towers (Liu et al. 2022, 2024a; 2024b, 2025), comprehensive studies addressing bridge components subjected to MS–AS sequences in subduction zones remain scarce. Contributions include Hidden Markov models for girder bridges (Chen et al. 2024), deterioration-focused fragility assessments (Otárola et al. 2022), and resilience-based evaluations (Gidaris et al. 2022). These works show that ASs amplify damage, reduce capacity, and alter functionality, but rarely address the influence of IM selection on predictive capability in MS–AS contexts.

The choice of IMs has been widely studied for single-event fragility but rarely for sequential loading. Scalar IMs such as PGA and $Sa(T_1)$ have well-documented limitations in capturing cumulative damage (O'Reilly 2021b; Yu et al. 2021). Alternatives—including peak ground velocity (PGV), average spectral acceleration ($Sa_{avg}(T_1)$, and vector- or energy-based IMs—show improved efficiency and sufficiency for some bridge types (Zhong et al. 2016; Franco et al. 2025). However, even though traditional IMs like Housner Intensity (HI) have been extensively used, their comparative optimality specifically for subduction zone MS–AS sequences remains unverified. Consequently, this study does not propose new IMs but rigorously evaluates the efficiency of these existing measures in this novel context.

The role of ground-motion duration and cumulative damage is also central in MS–AS assessment, because sequential shaking extends the effective duration of excitation and increases hysteretic energy demand. Duration has been shown to influence cumulative damage measures and collapse capacity of steel and reinforced-concrete systems (Hancock and Bommer 2006; Raghunandan and Liel 2013; Chandramohan et al. 2016; Barbosa et al. 2017), and duration-related measures such as significant duration provide complementary information on the temporal distribution of input energy (Trifunac and Brady 1975). In parallel, damage indices such as the Park–Ang formulation explicitly combine deformation and hysteretic-energy terms, making them well suited to quantify cumulative damage in reinforced-concrete components under extended or repeated shaking (Park and Ang 1985; Cosenza et al. 1993; Teran-Gilmore and Jirsa 2005). Regarding IM selection, previous studies on buildings and bridges have shown that velocity- and energy-based measures such as

HI and PGV (Wang et al. 2018; Jiang et al. 2019), spectrum-averaged measures (Eads et al. 2015; Du et al. 2019), and multi-metric evaluation approaches (Padgett et al. 2008; Gehl et al. 2013) can outperform conventional peak-based IMs depending on the structural system, the response quantity of interest, and the duration content of the ground motions (Fayaz et al. 2020). Nevertheless, such approaches remain rarely adapted to MS–AS contexts in subduction zones.

The limited bridge-specific MS–AS fragility studies and lack of comparative IM evaluation represent a critical research gap. Existing approaches either assess MS–AS without optimizing IM selection (Chen et al. 2022; Otárola et al. 2022) or evaluate IMs without modelling sequential demands (Lei et al. 2021; Zhou et al. 2024). To the authors' knowledge, no prior study has systematically compared candidate IMs for bridges under MS–AS sequences in subduction environments, underscoring the need for probabilistic frameworks that integrate realistic MS–AS hazard characterization, hazard-consistent GM selection, and systematic IM evaluation at the component level.

Experimental and numerical studies show that AS vulnerability is component-dependent: bearings and abutments tend to exhibit increased susceptibility during ASs due to MS-induced degradation, whereas columns are typically governed by MS demands (Yu et al. 2021; Afsar Dizaj et al. 2022). Reported increases in failure probability from MS–AS loading range from 20 to 40% relative to MS-only scenarios (Di Trapani and Malavisi 2019; Otárola et al. 2022). These findings highlight the need for component-level fragility assessments that incorporate cumulative effects and are supported by IMs capable of capturing these patterns.

Most MS–AS fragility assessments still rely on scalar IMs designed for single-event applications (Baker and Cornell 2006). Although vector- and energy-based IMs show promise (Fayaz and Galasso 2022; Fayaz et al. 2023), their adoption in subduction contexts is limited due to lack of calibration and integration into hazard-consistent frameworks.

Building on the preceding background, this study evaluates existing scalar, spectral, integral, duration-based, and orientation-independent IMs to identify component-optimal predictors of bridge demand under hazard-consistent MS–AS sequences in a subduction environment. The contribution is threefold: (i) the integration of stochastic MS–AS sequence characterization, conditional IM generation through a recurrent-neural-network ground-motion model, and multi-objective hazard-consistent record selection within a consistent IM-evaluation workflow; (ii) the systematic component-level assessment of candidate IMs using four complementary criteria—efficiency, practicality, proficiency, and sufficiency; and (iii) component-specific optimal-IM recommendations for elastomeric bearings, abutment backfill, and reinforced-concrete columns, demonstrated on the Águila Norte Bridge, an existing RC highway bridge in central Chile. The remainder of the paper is organized as follows: Sect. 2 presents the methodological framework; Sect. 3 describes its application to the Águila Norte Bridge; Sect. 4 reports the IM-evaluation results; and Sect. 5 summarizes the conclusions and their implications for MS–AS fragility assessment.

2 Methodological framework

2.1 Overall description of the proposal

Figure 1 presents a schematic overview of the methodological framework designed to identify component-optimal IMs for bridges subjected to MS–AS sequences. The framework comprises three interdependent modules: (i) probabilistic characterization of site-specific MS–AS scenarios and definition of conditional target IM vectors; (ii) nonlinear time-history analysis (NLTHA) of the bridge under hazard-consistent MS–AS records with extraction of component-level engineering demand parameters (EDPs); and (iii) comparative IM evaluation based on efficiency, practicality, proficiency, and sufficiency.

Hazard characterization starts with an APSHA that defines site-specific MS–AS scenarios across selected hazard levels (HLs). A recurrent-neural-network ground-motion model (RNN-GMM) provides conditional target IM vectors, and a multi-objective record selection algorithm assembles hazard-consistent MS–AS sequences. Structural response is then evaluated through NLTHA of a finite element (FE) bridge model, capturing effects on elastomeric bearings (EB), abutment backfill (BF), and RC columns (COL). Finally, probabilistic seismic demand models (PSDMs) relate EDPs to candidate IMs, which are compared across four metrics: efficiency, practicality, proficiency, and sufficiency. Practicality, understood

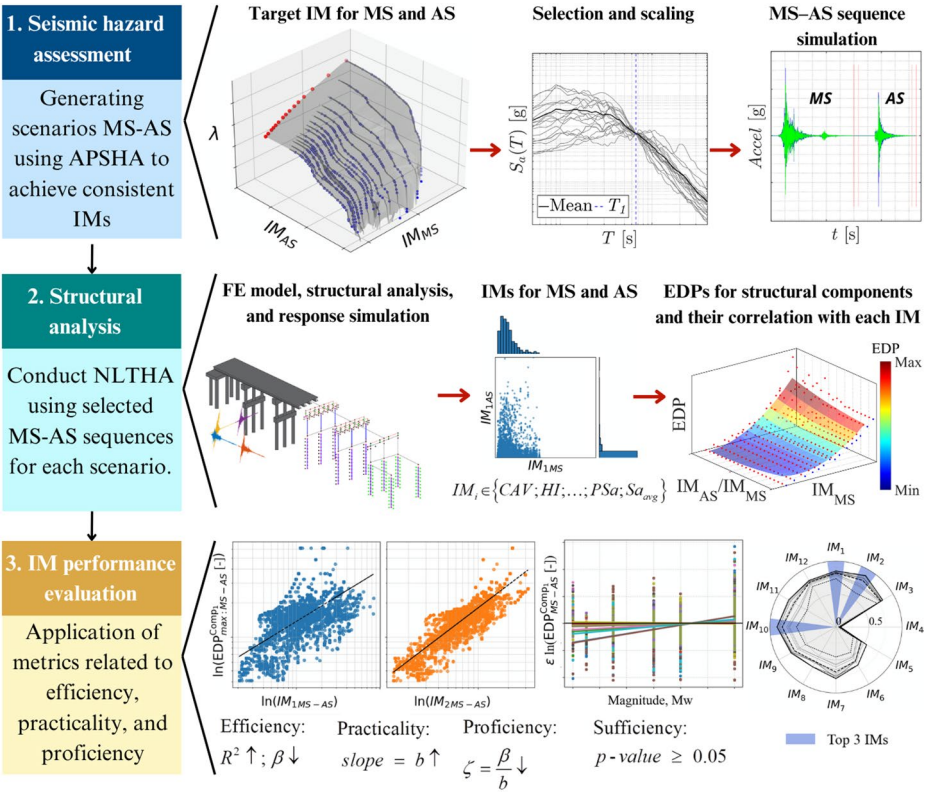

Fig. 1 Schematic framework for optimal IM evaluation under MS–AS sequences

here as structural-demand sensitivity rather than hazard computability, is represented by the regression slope b, where larger *b* values denote stronger proportional demand variation per IM decade. The framework ultimately identifies the most suitable IM predictors for fragility assessment (see Sect. 2.5 for full details).

2.2 Probabilistic MS–AS input characterization

This study adopts the APSHA framework, originally proposed by (Cagua et al. 2026), which extends the classical PSHA formulation (Cornell 1968) to explicitly model AS occurrence and provide hazard-consistent IM pairs while preserving inter-event dependence in magnitude, distance, and spectral features. In that companion study, the framework was developed and applied to MS–AS fragility assessment of a Chilean highway bridge using $Sa(T_1)$ as the primary conditioning IM. The present manuscript adopts the same hazard-consistent MS–AS input characterization but addresses a distinct objective: the systematic evaluation of existing IMs for component-level bridge fragility under MS–AS sequences. The procedure comprises the following four main stages.

i) PSHA and disaggregation

A conventional PSHA computes the mean annual rate of exceedance (MRE), denoted by $\lambda(\mathrm{IM} > im)$, for a selected IM (e.g., $Sa(T_1)$). For each HL with return period (*Tr*), seismic disaggregation yields representative MS scenarios $w \in \{1, \ldots, n_{Tr}\}$ defined by triplet magnitude, rupture distance, and hypocentral depth of the MS, denoted by M_{MS}, R_{MS}, Z_{MS}, respectively.

ii) Stochastic generation of AS scenarios

For each disaggregated MS scenario, a set of consistent AS events $z \in \{1, \ldots, n_{AS^w}\}$ is generated using the Epidemic-Type Aftershock Sequence (ETAS) model (Ogata 1998; Ogata and Zhuang 2006), calibrated for subduction zones. The model incorporates key empirical laws—Gutenberg–Richter recurrence (Gutenberg and Richter 1944), Bath's law (Saichev and Sornette 2005), and the Omori–Utsu decay model (Omori 1895; Utsu et al. 1995)—to capture realistic AS occurrence and distribution.

Each AS is conditioned on the parent MS and defined by four parameters: magnitude, rupture distance, and depth of the AS (M_{AS}, R_{AS}, and Z_{AS}, respectively) and rupture dip angle α. These are sampled through region-specific stochastic formulations driven by uniformly distributed random variables u_{m1}, u_{m2}, u_{m3}, $u_{m4} \sim \mathrm{U}(0,1)$, where $\mathrm{U}(a,b)$ denotes a uniform distribution in the range $[a,b]$. The governing equations are:

$$M_{AS_z^w} = -\frac{ln\,(1 - u_{m1})}{m \cdot ln\,(10)} + M_{min}; \qquad M_{AS_z^w} < M_{MS_w} \tag{1}$$

$$Z_{AS_z^w} = Z_{min} + (Z_{max} - Z_{min}) \cdot u_{m2}\ [km] \tag{2}$$

$$\alpha_z = \alpha_{min} + (\alpha_{max} - \alpha_{min}) \cdot u_{m4} \qquad [^\circ] \tag{3}$$

$$D_{MS_w - AS_z^w} = d \cdot \; e^{\gamma\,(M_{MS_w} - M_{\min})} \cdot \; \sqrt{u_{m3}^{\frac{1}{1-q}} - 1} \;\; [\mathrm{km}] \tag{4}$$

$$R_{AS_z^w} = R_{MS_w} - D_{MS_w - AS_z^w} \cdot \; sin\,(\alpha_z) \;\; [km] \tag{5}$$

where m is the Gutenberg–Richter slope, M_{min} the minimum M considered for the region, d, γ, q empirical constants, and [Z_{min}, Z_{max}], [α_{min}, α_{max}] the depth and dip ranges reflecting the local tectonic regime (e.g., 10°–25° for subduction), respectively.

In Eq. 4, d is a spatial scale parameter [km] governing the characteristic MS–AS separation, γ controls the magnitude-dependent spread of AS distances, and q is a shape parameter governing the heaviness of the spatial-decay tail. These parameters must be calibrated to the seismotectonic context of the study area; the values adopted for the Chilean subduction environment are reported in Sect. 3.2.

Physical plausibility is enforced through three constraints: $R_{AS_z^w} \geq R_{min}$ to avoid near-source saturation, $\left|R_{AS_z^w} - R_{MS_w}\right| \leq \Delta R$ to preserve clustering, and location within regionally observed bounds (Z_{AS} and α restricted to [Z_{min}, Z_{max}] and [α_{min}, α_{max}], as defined in Eqs. 2 and 3). All parameters must be consistent with the seismotectonic setting and validated against historical catalogs. The calibrated values adopted for the Chilean subduction environment are reported in Sect. 3.2.

iii) Conditional IM for MS-AS

To ensure physically consistent seismic IMs between MS and AS events, a RNN-GMM is employed. This model leverages the sequential nature of MS–AS pairs to generate conditional IM vectors that preserve the amplitude scaling, spectral shape, and duration dependencies inherent to real seismic sequences as demonstrated in (Fayaz and Galasso 2022). In the first stage, the model predicts the IM vector for each MS scenario as a function of key seismological and site parameters:

$$\mathrm{IM_{MS}} = f_{\mathrm{RNN}}\,(M_{\mathrm{MS}}, R_{\mathrm{MS}}, Z_{\mathrm{MS}}, Vs_{30}\;,\; F) \tag{6}$$

where Vs_{30} is the time-average shear-wave velocity of the upper 30 m at the site and F refers to the tectonic classification (e.g., subduction interface, intraslab, crustal).

In the second stage, the predicted MS IMs are used to generate the corresponding AS IMs, conditioned on both MS outputs and AS characteristics:

$$\mathrm{IM_{AS}} = f_{\mathrm{RNN}}\,(IM_{MS},\; \mathrm{M_{AS}}, R_{AS}, Z_{AS}) \tag{7}$$

This sequential architecture captures the dependence of AS intensities on the preceding MS, as well as on the M and location of the AS event itself. The output is a joint IM vector:

$$\mathbf{t} = \begin{bmatrix} IM_{MS} \\ IM_{AS} \end{bmatrix} \tag{8}$$

which includes multiple IMs relevant for fragility assessment, such as $Sa(T_1)$, PGV, cumulative absolute velocity (CAV), and GM duration metrics.

iv) Joint hazard surface

Conditional IM vectors are integrated into a joint hazard formulation based on the APSHA framework. The MRE for each paired IM threshold is expressed as in (Yu et al. 2024):

$$\lambda\ (IM_{AS} > v, IM_{MS} > y) = \sum\nolimits_{w=1}^{n} P \begin{pmatrix} (IM_{AS} > v \mid\ M_{AS}, R_{AS}) \\ \mid\ (IM_{MS} > y \mid\ (M_{MS}, R_{MS})_w) \\ \cdot \lambda\ (IM_{MS} > y \mid\ (M_{MS}, R_{MS})_w) \end{pmatrix} \quad (9)$$

In Eq. (9), $\lambda(IM_{AS} > v, IM_{MS} > y)$ denotes the joint exceedance rate of simultaneously surpassing thresholds v and y during the AS and MS, respectively. Each pair $(M_{MS}, R_{MS})_w$ denotes the w^{th} MS scenario obtained from PSHA disaggregation. The term $\lambda(IM_{MS} > y \mid (M_{MS}, R_{MS})_w)$ is the conditional exceedance rate of the MS IM, while the probability term $P[\cdot]$ quantifies the likelihood that the AS IM exceeds threshold v given the MS scenario and stochastically generated AS parameters (M_{AS}, R_{AS}).

2.3 Ground motion selection and seismic intensity measures

The conditional target IM vectors $[IM_{MS}, IM_{AS}]$ are defined for each HL using the probabilistic framework and the RNN-GMM approach detailed in Sect. 2.2. A multivariate GM selection process is then carried out. GMs corresponding to MS and AS events are retrieved from curated databases (e.g. (Aoi et al. 2004; Castro et al. 2022), with each event type classified independently to ensure physical consistency between the selected records and their intended role in the sequence. GM selection is conducted through a multicriteria optimization procedure that minimizes the discrepancy between scaled IMs of candidate records and the previously defined target vectors. The objective function integrates several IMs—$Sa(T_1)$, PGV, CAV, Arias intensity (Ia), and significant duration (DS_{5-95})—to ensure compatibility in both spectral content and energy-related characteristics. Scaling factors are suggested to remain below a maximum value of 4.0 to prevent unrealistic modifications to the original GMs. This selection strategy builds upon the methodology introduced in (Fayaz and Galasso 2022) and implemented in (Fayaz 2023).

MS and AS records are scaled independently to their respective targets, and horizontal components are combined using the geometric mean, consistent with the APSHA-based definition of $Sa(T_1)$. While this involves uncoupled scaling factors for the MS and AS components, this approach is justified because the physical inter-event correlation is enforced at the hazard definition stage by the RNN-GMM-derived target vectors (Sect. 2.2), not by the scaling process itself. Crucially, the RNN-GMM ensures that the generated AS intensity is not random but conditionally dependent on the MS history, preserving the spectral shape correlations observed in real subduction records. Each scaled pair is merged into a single input motion for NLTHA, with a free vibration time gap of $25T_1$ introduced between MS and AS records to allow the MS-induced transient response to decay and re-establish near-equilibrium conditions prior to the AS, consistent with established practice (Wang et al.

2022; Chen et al. 2022, 2024), followed by an identical gap to capture the residual post-AS response.

To assess correlations between seismic demand and structural response, a comprehensive set of IMs is computed from each selected sequence (Table 1). The mathematical functions for these IMs, detailed in the table, utilize several standard seismological variables. Specifically, $a(t)$ denotes the ground acceleration time-history, $V(t)$ is the ground velocity time-history, and t_f represents the total duration of the ground motion record. For spectral parameters, T is the natural period of vibration, ξ is the viscous damping ratio (typically assumed as 5%), $PSV(\xi, T)$ is the pseudo-spectral velocity, $Sa(T, \xi)$ is the pseudo-spectral acceleration, often written simply as $Sa(T)$ for a fixed damping ratio, and $S_d(T, \xi)$ is the spectral displacement. Ia denotes the Arias Intensity (used to compute DS_{5-95}), ν_0 is the rate of zero crossings of the acceleration record, DP is the Destructive Potential, and t_5 and t_{95} are the instants at which the cumulative Ia reaches 5% and 95% of its total value, and g is the acceleration due to gravity. For rotational measures, $a_x(t)$ and $a_y(t)$ represent the acceleration time-histories of the two orthogonal horizontal components, and ω is the rotation angle; the orientation-independent measures $Sa_{RotD50}(T_1)$ and $Sa_{RotD100}(T_1)$ are defined as the median and maximum of the spectral response over all angles ω.

The selected IMs include amplitude-, energy-, and duration-based measures such as $Sa(T_1)$, PGV, CAV, and DS_{5-95}. The average spectral acceleration $Sa_{avg}(T_1)$ is calculated over the period range $0.2T_1$–$3T_1$, following established practice for spectrum-averaged IMs (Eads et al. 2015): the lower bound incorporates higher-mode and stiffening contributions, while the upper bound accommodates the period elongation induced by stiffness degradation under cumulative MS–AS loading. It is acknowledged that displacement-governed components, such as the abutment backfill, may exhibit sensitivity to spectral ordinates outside this range, which can influence the relative ranking of $Sa_{avg}(T_1)$ for those components. In its expression, c_i are the factors defining the N periods c_iT_1 within this range. The velocity pulse index ($FIV3(T_1)$) is extracted from filtered velocity time series using a second-order Butterworth filter (cutoff = 1.0 Hz) and $0.7T_1$ time windows. The incremental-velocity series $V_s(t)$ is computed over segments of duration ΦT_1, where ($\Phi = 0.7$) is a dimensionless integration-window coefficient; $V_{s,\,max\,i}$ and $V_{s,\,min\,i}$ ($i = 1,2,3$) denote the first three local maxima and minima of $V_s(t)$; $\ddot{u}_{gf}$ is the filtered ground acceleration; and t_{end} is the final instant of the record and ψ is the integration time variable. This approach produces hazard-consistent seismic sequences that accurately reflect the compound loading effects of MS–AS events, ensuring both physical realism and methodological robustness for probabilistic fragility assessment.

Each IM is evaluated under four representations of seismic input:

- IM: MS – computed from the MS record alone, enabling comparison with conventional single-event fragility models.
- IM: max(MS, AS) – the maximum of MS and AS values, capturing peak demand regardless of event order.
- IM: MS–AS (concatenated) – computed from the full MS–AS sequence, reflecting cumulative and duration-sensitive effects.
- IM: MS, AS (independent) – computed separately for MS and AS and stored as paired vectors, supporting multivariate correlation with EDPs.

Table 1 Ground motion parameters

No.	Ground Motion Parameter	Mathematical Function	Ref.
1	Cumulative absolute velocity (CAV)	$\mathrm{CAV} = \int_0^{tf} \lvert a(t)\rvert \, dt$	(Reed and Kassawara 1990)
2	Significant duration (DS_{5-95})	$DS_{5-95} = t_{95} - t_5$	(Trifunac and Brady 1975)
3	Housner intensity (HI)	$\mathrm{HI}(\xi) = \int_{0.1}^{2.5} PSV(\xi, T) dT$	(Housner 1952)
4	Arias intensity (Ia)	$Ia = \frac{\pi}{2g} \int_0^{t_f} [a(t)]^2 dt$	(Arias 1970)
5	Destructive potential (DP)	$\mathrm{DP} = \frac{Ia}{v_0^2}$	(Araya and Saragoni 1984)
6	Peak ground acceleration (PGA)	$\mathrm{PGA} = max\,(Abs(\,a\,(t)\,))$	(Gutenberg and Richter 1942)
7	Peak ground velocity (PGV)	$\mathrm{PGV} = max\,(Abs(\,V\,(t)\,))$	(Newmark and Hall 1973)
8	Geometric mean pseudoacceleration response spectra ($Sa(T)$)	$Sa(T, \xi) = \frac{4\pi^2}{T^2} \cdot S_d(T, \xi)$	(Newmark and Hall 1982)
9	Average acceleration spectra ($Sa_{avg}(T)$)	$Sa_{avg}\,(c_1 T_1, \ldots, c_N T_1) = \left(\prod_{i=1}^{N} Sa\,(c_i T_1)\right)^{1/N}$	(Eads et al. 2015)
10	Filtered incremental velocity ($FIV3(T)$)	$FIV3\,(T_1) = max\,\{V_{s,max1} + V_{s,max2} + V_{s,max3}, \lvert V_{s,min1} + V_{s,min2} + V_{s,min3}\rvert\}$ $V_s \mid (t) = \left\{\int_t^{t+\Phi \cdot T_1} \ddot{u}_{gf}\,(\psi)\, d\psi, \forall\, t < t_{\mathrm{end}} - \Phi \cdot T_1\right\}$	(Dávalos and Miranda 2020)
11	Acceleration response spectra $Sa_{RotD50}(T)$, $Sa_{RotD100}(T)$	$a_z(t, \omega) = a_x(t) \cdot cos\,(\omega) + a_y(t) \cdot sin\,(\omega)$ $Sa_{\mathrm{RotD50}}\,(T) = median\,(Sa\,(T, \xi, \omega))$ $Sa_{\mathrm{RotD100}}\,(T) = max\,(Sa(\,T, \xi, \omega\,))$	(Boore et al. 2006)

Notation: all symbols are defined in Sect. 2.3; spectral ordinates are evaluated using 5% viscous damping (ξ = 5%); $Sa_{avg}(T_1)$ is averaged over $0.2T_1$–$3T_1$, and FIV3(T_1) uses a second-order Butterworth filter (1.0 Hz cutoff) and time windows of duration $0.7{\cdot}T_1$

In addition to the four primary IM representations, an auxiliary scalar combination is evaluated for comparison purposes: the square root of the sum of squares, SRSS(MS, AS) = $\sqrt{(IM_{MS}^2 + IM_{AS}^2)}$. This representation combines the MS and AS intensity levels into a single scalar predictor and is used solely to compare alternative MS–AS scalar-combination operators in the two-dimensional IM–EDP efficiency assessment.

2.4 Structural modeling and nonlinear seismic analysis

This section describes the general structural-modeling requirements of the proposed IM-evaluation workflow. The specific OpenSees implementation adopted for the Águila Norte Bridge—including the modeled components, constitutive laws, boundary conditions, analysis settings, and extracted EDPs—is detailed in Sect. 3.1.

To evaluate the structural response under MS–AS seismic sequences, a nonlinear FE model is developed using a computational platform capable of performing NLTHA. The model incorporates material and geometric nonlinearities, as well as appropriate constitutive laws to capture stiffness and strength degradation, cyclic behavior, and residual deformations. Software platforms such as OpenSees (McKenna et al. 2000) are particularly suited for this task, given their proven capability in simulating complex dynamic behavior under sequential earthquake loading. The structural model accurately represents the geometry, boundary conditions, and dynamic properties of the bridge system under study. Key components—such as COL, EB, and BF—must be explicitly modeled with sufficient resolution to track damage-sensitive response parameters. Where applicable, soil–structure interaction effects, mass eccentricities, and energy dissipation mechanisms (e.g., isolation, damping devices) should be included.

Each selected MS–AS GM sequence is applied to the model to perform NLTHA. The simulation captures the full duration of the seismic input, including the inter-event interval (to re-establish equilibrium) and post-AS residual phase. Given the computational demand, analyses were executed in parallel using high-performance computing (HPC) resources. Parallel batch execution across HLs enables efficient simulation and systematic data extraction across multiple seismic scenarios.

A range of EDPs were computed from the structural response histories, including maximum relative displacements of EB, cumulative damage indices for COL, and relative deck–backwall displacements for BF.

In addition to these standard EDPs, energy-based metrics such as hysteretic energy dissipation may also be computed to evaluate cumulative damage and degradation. These EDPs are selected to reflect component-specific vulnerability mechanisms and to provide robust input for subsequent fragility analysis.

Time-history responses are post-processed to extract peak EDP values at three instants: (i) the maximum during the MS, (ii) the maximum during the AS, and (iii) the overall maximum across the full MS–AS sequence (i.e., the greater of the two). This systematic segmentation enables evaluation of damage progression and residual capacity, while providing a consistent dataset for developing fragility curves under MS-only, and sequential MS–AS scenarios.

The segmented EDPs extracted from the NLTHA are paired with the IM representations defined in Sect. 2.3 to construct the IM–EDP datasets used in the statistical evaluation. Five pairings are considered: (i) IM_{MS} with the maximum EDP during the MS; (ii) max(IM_{MS},

IM_{AS}) with the maximum EDP over the complete MS–AS sequence; (iii) IM_{MS-AS}, computed from the concatenated sequence, with the maximum sequence EDP; (iv) the paired vector (IM_{MS}, IM_{AS}) with the maximum sequence EDP; and (v) the auxiliary IM_{SRSS} with the maximum sequence EDP. These pairings provide the basis for the subsequent PSDM fitting and IM-performance evaluation.

2.5 Evaluation criteria for optimal intensity measures

Optimal IMs for probabilistic fragility assessment are identified using four evaluation criteria: efficiency, practicality, proficiency, and sufficiency. These are formulated through the PSDM:

$$\ln(\mathrm{EDP}) = a + b \cdot \ln(\mathrm{IM}) + \varepsilon\,, \qquad \varepsilon \sim \mathcal{N}\left(0, \beta^{2}\right) \tag{10}$$

where EDP are obtained from NLTHA, a and b are regression coefficients, and β is the conditional logarithmic standard deviation, and ε denotes the residual term, which is assumed to follow a normal distribution ($\mathcal{N}$) with zero mean and variance β^2.

Efficiency quantifies the dispersion of demand estimates for a given IM. A more efficient IM produces lower residual variability β, indicating greater predictive power. The coefficient of determination, R^2, is also used to support this assessment, with higher values implying stronger correlation between ln(EDP) and ln(IM) (Luco and Cornell 2007).

In this paper, Practicality is defined as the sensitivity of the structural response to the selected IM, rather than as hazard computability or ease of IM implementation within PSHA, and is quantified by the regression slope b from Eq. (10). This slope reflects the rate of change in logarithmic demand relative to the logarithmic IM. A statistically significant and non-zero slope b indicates a practical IM, as it confirms that the IM contributes meaningfully to explaining the observed demand (Mackie and Stojadinović 2001).

Proficiency is a composite metric that balances efficiency and practicality. It is defined as:

$$\zeta = \frac{\beta}{b} \tag{11}$$

A lower proficiency index ζ suggests that the IM yields stable predictions with relatively low dispersion and high sensitivity. This metric facilitates direct comparison between IMs, especially when efficiency and practicality trade off (Padgett et al. 2008). It is noted, however, that this index can become numerically unstable if the practicality slope is very close to zero; thus, it is most meaningful when comparing IMs that have already demonstrated significant practicality.

Sufficiency evaluates the statistical independence of the PSDM residuals ε from key GM parameters, particularly M and R. An IM is considered sufficient if these residuals show no significant trends with respect to M or R. This is tested via correlation; sufficiency is accepted when the null hypothesis of independence is not rejected (e.g., $p \geq 0.05$) (Luco and Cornell 2007).

2.6 Model for mainshock–aftershock exceedance probability surfaces

The joint influence of MS and AS intensities on structural performance is quantified using a bivariate logistic regression model that estimates the conditional probability of exceeding a given damage state (DS), similar to that adopted in (Kourehpaz et al. 2023). This approach is applicable to any structural component—such as COL or EB—and enables the construction of exceedance probability surfaces under sequential seismic input.

A binary response variable is defined to indicate whether a selected EDP surpasses the prescribed DS threshold. The predictors are the natural logarithms of two seismic IMs, one for the MS and one for the AS. To capture nonlinear effects and MS–AS interactions, the model includes quadratic and cross-product terms, yielding the following functional form:

$$\begin{aligned}\text{logit}\,(P) = b_0 + b_1 \ln\left(IM_{MS}\right) + b_2 \ln\left(IM_{AS}\right) + b_3 \ln(IM_{MS})^2 \\ + b_4 \ln(IM_{AS})^2 + b_5 \ln\left(IM_{MS}\right) \ln\left(IM_{AS}\right)\end{aligned} \tag{12}$$

where P denotes the probability of exceeding a given DS, and b_0-b_5 are regression coefficients.

3 Application to Águila Norte Bridge

3.1 Structural system and numerical modeling

The Águila Norte (AN) bridge is a RC highway overpass with five simply supported spans and a total length of 148 m. Each span measures nearly 29.6 m, and the 10.7 m-wide deck consists of precast, post-tensioned I-girders with a 20 cm cast-in-place RC slab, connected transversely by RC diaphragms (Pasten et al. 2021). The substructure comprises four three-column bents and two seat-type BF, supported on 1.2 m-diameter drilled shafts embedded 10–16.8 m into dense gravel deposits (VS_{30} = 277 m/s, Site Class II), with a site period of 0.5 Hz (Guajardo et al. 2024).

To safeguard the structure under seismic excitation, the bridge is equipped with EB, longitudinal restrainers, and external shear keys at all support interfaces. These components function collectively to restrict excessive movement, prevent unseating, and absorb seismic energy. During the 2010 Maule earthquake (moment magnitude, *Mw 8*.8), the bridge exhibited limited damage, attributed to its ductile detailing and overall balanced seismic performance.

Figure 2 provides a visual summary of the bridge configuration, pier geometry, and fiber-section modeling strategy. The numerical model of the bridge was developed using OpenSees (McKenna et al. 2000). While Soil-Structure Interaction (SSI) can alter demand, a fixed-base assumption was adopted here to isolate the MS–AS sequence effects on component capacity. The bridge deck was modeled with elastic Timoshenko beam-column elements to account for shear deformation effects. Cap beams and piles were also modeled as elastic elements, with rigid offsets used at the interfaces to replicate the continuity of cast-in-place connections. Diaphragm constraints were enforced at the top of each pier cap to emulate the rigid transverse behavior of the superstructure. COLs were discretized using force-based beam-column elements to simulate distributed plasticity and capture nonlinear

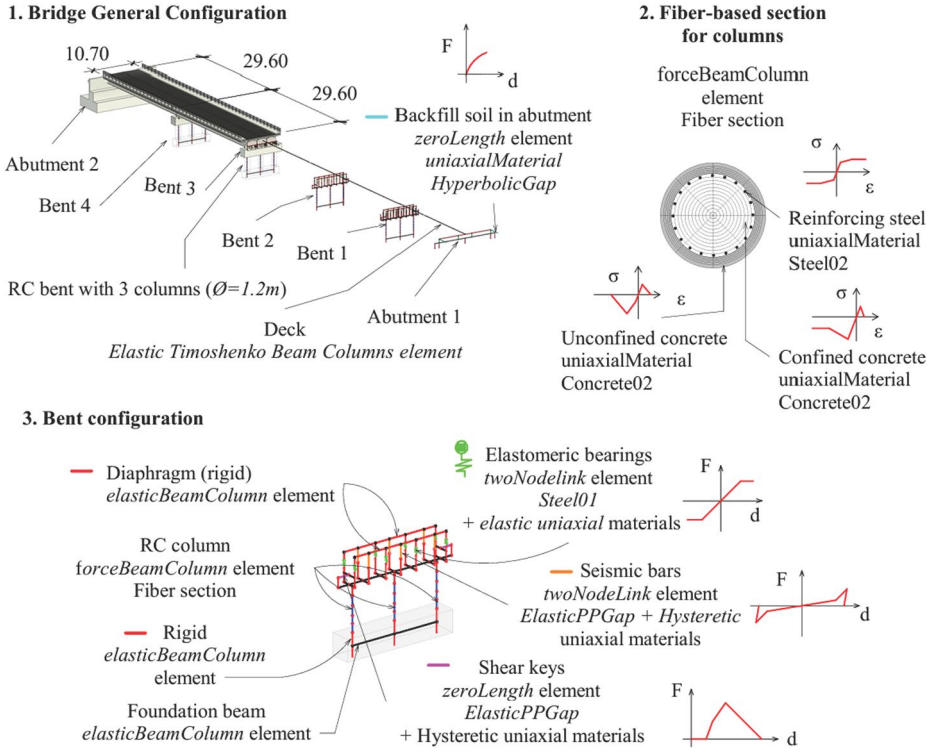

Fig. 2 FE model of the AN bridge

flexural behavior. Each COL, 1.2 m in diameter, incorporates 20 longitudinal bars of 28 mm and transverse hoops of 12 mm spaced at 50 mm. The confined concrete core and unconfined cover were represented using the *Concrete02* material model, while steel reinforcement was modeled with *Steel02*, enabling the capture of Bauschinger effects and cyclic degradation. EBs were simulated using the *Steel01* model in the horizontal direction (Rubilar Moya 2016), while their vertical and rotational stiffnesses were modeled as linear elastic. Seismic bars were implemented as a combination of a bilinear *Hysteretic* material for lateral stiffness and an *ElasticPPGap* element for vertical restraint (Martínez Grosser 2016). Exterior shear keys were modeled as *Hysteretic* components in series with *ElasticPPGap* elements to account for nonlinearity (Goel and Chopra 2008). The BF resistance was incorporated via the *HyperbolicGap* material model, calibrated against empirical stress–displacement curves for compacted granular soils (Duncan and Mokwa 2001).

3.2 Seismic hazard characterization and scenario definition

Fifteen HLs with *Tr* from 50 to 3000 years were defined using PSHA considering only MS contributions (computed using the SeismicHazard platform (Candia et al. 2019), based on SLAB2 geometry (Hayes et al. 2018), Chilean zoning (Poulos et al. 2018), and a logic tree with five subduction GMMs (Montalva et al. 2017; Idini et al. 2017; Abrahamson and Gülerce 2020; Kuehn and Abrahamson 2020; Parker et al. 2021) with equal weights. Con-

ditional disaggregation yielded representative MS scenarios in terms of M, R, and Z, conditioned on the fundamental period of the bridge ($T_1 = 0.66$ s). For each HL, physically consistent AS events were stochastically generated with a Chile-calibrated ETAS model (Nicolis et al. 2015), enforcing spatial ($\Delta R \leq 25$ km, $R_{min} \geq 10$ km) and dip-angle (10°–25°) constraints. Calibrating ETAS parameters specifically to the Chilean catalog ensures that the triggering rates and magnitude distributions reflect the high productivity of ASs typical of this subduction environment. From 10000 AS realizations per HL, 40 representative scenarios were retained, combining kernel-density sampling with upper-percentile cases (90th, 95th, and 99th) to ensure both representativeness and coverage of extremes.

AS scenarios exhibit greater variability, being more distant and moderate at low HLs but clustering closer to the MS source with slightly larger M at high HLs, thus amplifying cumulative demand. In terms of IMs, MS $Sa(T_1)$ values grow steadily with Tr, exceeding 2.0 g in extreme cases, while ASs frequently surpass 1.0 g and occasionally approach 1.5 g. Further details on the statistical behavior of MS–AS scenarios are provided in (Cagua et al. 2026).

To quantify the joint occurrence of $Sa(T_1)$ for MS and AS events, a hazard surface was constructed for the pair ($Sa(T_1)_{\mathrm{MS}}$, $Sa(T_1)_{\mathrm{AS}}$). The selected IM ($Sa(T_1)$) reflects both design relevance and structural sensitivity for the bridge. The surface was discretized across selected Tr (e.g., 100, 475, 950, 2475 years) to support Multi-Stripe Analysis (MSA). For each HL, a target vector $\mathbf{t}$ was defined as the conditional mean of $[IM_{MS}, IM_{AS}]$, providing hazard-consistent input for record selection and NLTHA.

Figure 3 illustrates the hazard surface for the AN Bridge in central Chile, showing monotonic growth along both axes and encompassing the full range of stochastically generated

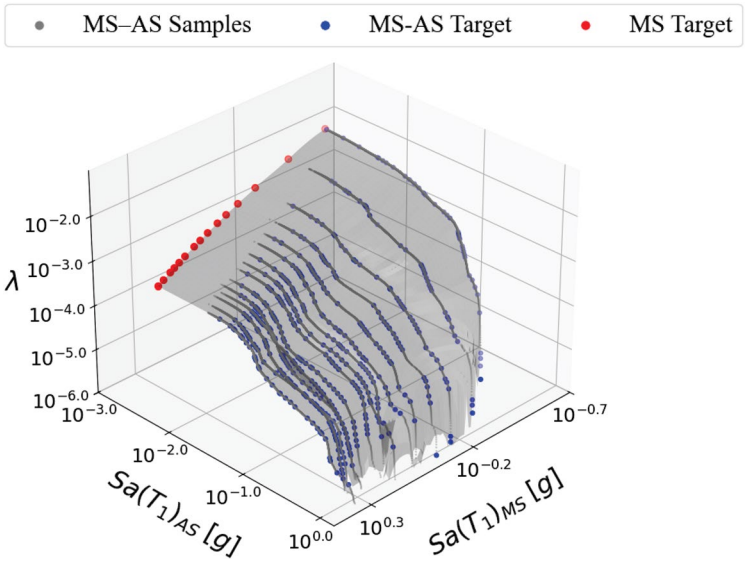

Fig. 3 Seismic hazard surface for MS-AS sequences

realizations (gray cloud). Red markers denote marginal MS targets for each HL, while blue markers represent conditionally derived AS targets. Their alignment with the high-density cloud confirms consistency across disaggregation, stochastic simulation, and joint hazard representation, providing a robust basis for IM-based record selection.

To evaluate structural performance under MS–AS sequences, 12,000 NLTHA were performed across 15 HLs. For each HL, a representative MS scenario was obtained through conditional disaggregation, and 40 physically consistent AS scenarios were selected. From this pool, 20 MS–AS record pairs per HL were selected, balancing statistical representativeness with computational efficiency. Each pair combined a unique MS record and a corresponding AS record, ensuring seismological coherence in terms of source-to-site parameters and intensity scaling.

Figure 4 illustrates the relationship between MS and AS IMs from the NLTHA dataset and is retained as a dataset-level diagnostic of the seismic-intensity range considered in this study. Each subplot presents a bivariate comparison between an MS-based IM and its corresponding AS value, with marginal histograms showing the distributions. The plots reveal distinct correlation patterns depending on the IM type. Peak-based IMs (e.g., PGA, PGV) exhibit moderate positive correlation, with AS values typically lower but occasionally comparable to or exceeding MS levels, particularly at upper-percentile hazard thresholds. Energy- and duration-sensitive IMs (e.g., CAV, *Ia*, DS_{5-95}) show broader scatter and weaker trends, reflecting variability in shaking duration and cumulative energy. By contrast, spectral acceleration measures (e.g., $Sa(T_1)$, $Sa_{avg}(T_1)$, Sa_{RotD50}, $Sa_{RotD100}$) display stronger and more uniform correlation, reinforcing their potential as robust predictors of structural demand. Overall, the figure supports the need to evaluate multiple IM families before proceeding to the IM–EDP performance metrics.

Displacement-sensitive and integral IMs (e.g., HI, DP) present higher scatter, especially for AS events. This behavior is attributed to their nature as broad-band metrics that accumulate information throughout the ground motion duration. Specifically, HI integrates spectral velocity over a wide period range, allowing it to better capture the cumulative damage potential and structural period elongation effects inherent to MS–AS sequences, unlike peak-based metrics such as PGA. Similarly, DP is governed by the cumulative energy (*Ia*) of the full time-history, reflecting the progressive deterioration of the system. Overall, Fig. 4 underscores the need for a diverse set of IMs to fully capture the complex, energy-driven, and sometimes decoupled nature of seismic demand under MS–AS sequences.

Figure 5 shows the Pearson correlation matrix for the 12,000 MS–AS ground motion pairs, including both MS-only IMs (top block) and sequence-based IMs (bottom block). To improve interpretability, a discrete seven-band color scheme is adopted, distinguishing low, medium, and high correlations without overemphasizing sign. Spectral acceleration metrics (e.g., $Sa(T_1)$, $Sa_{avg}(T_1)$) exhibit high correlation ($\rho \geq 0.8$) across both groups, confirming their consistency and robustness as predictors of structural demand. Energy-based IMs such as *Ia* and CAV also display moderate cross-group correlation, indicating partial preservation of energy content across sequences. In contrast, duration-related IMs (DS_{5-95}) show weak correlations with the remaining IMs, reflecting their distinct statistical behavior and sensitivity to AS inclusion. PGA, PGV, and $FIV3(T_1)$ demonstrate moderate correlation (0.5–0.7), with greater stability in their MS–AS variants. Overall, the matrix highlights that while some IMs maintain predictive reliability when extended to full sequences, oth-

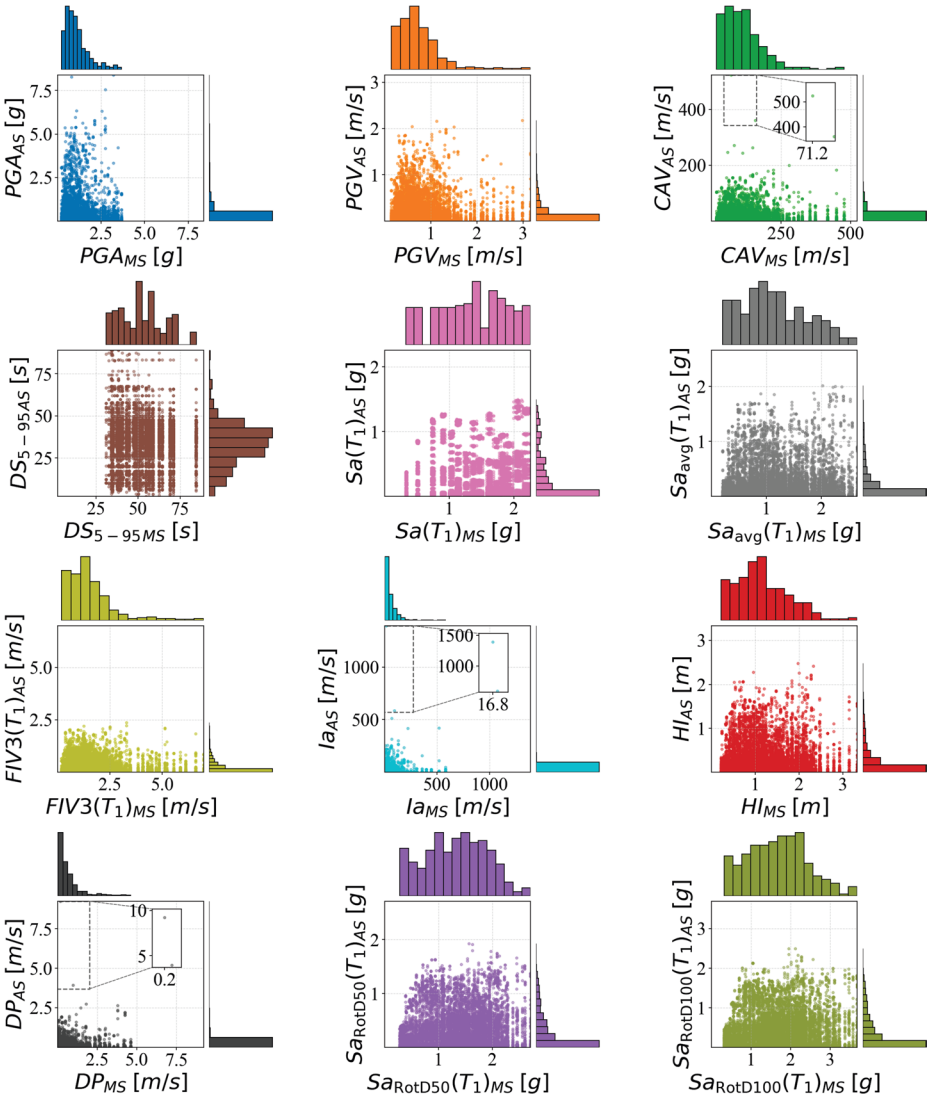

Fig. 4 Relationships between MS and AS IMs across the selected GM parameters

ers—particularly cumulative and duration-sensitive measures—are more strongly affected by AS effects.

Beyond these general trends, the matrix supports the consistency of the MS and AS generation and clarifies the role of duration. The strong correlations among spectral measures—$Sa(T_1)$, $Sa_{avg}(T_1)$, HI, and the rotated spectral accelerations, with $\rho > 0.95$ between their MS and MS–AS counterparts—confirm that the procedure preserves the spectral structure of the records throughout the sequence. The significant duration of the MS ($DS_{5-95\,MS}$) correlates only weakly with all

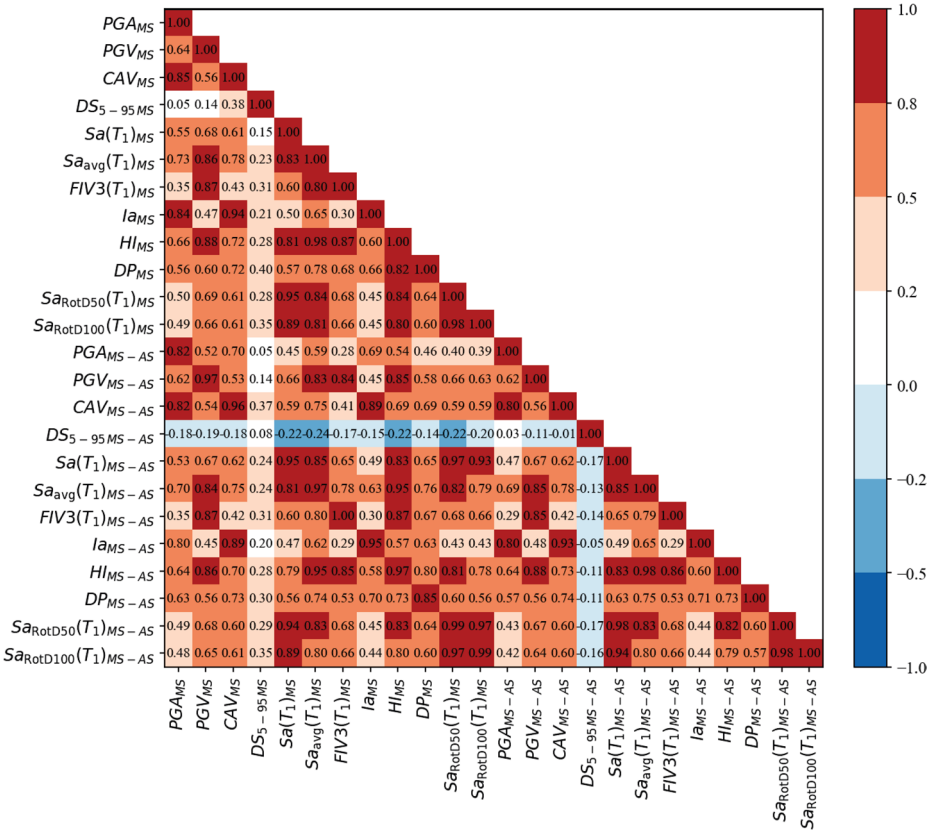

Fig. 5 Correlation Matrix Between MS and MS-AS IMs

other IMs ($|\rho| \leq 0.40$; e.g., 0.15 with $Sa(T_1)_{MS}$ and 0.28 with HI_{MS}), indicating that duration conveys information largely orthogonal to amplitude- and energy-based measures. Regarding sign, $DS_{5-95\,MS}$ and $DS_{5-95\,MS-AS}$ are essentially uncorrelated ($\rho = 0.08$), whereas the weak negative values in the figure (ρ between −0.13 and −0.24) correspond to the duration of the full sequence $DS_{5-95\,MS-AS}$ against the spectral intensity of the MS (e.g., −0.24 with $Sa_{avg}(T_1)_{MS}$). Physically, this indicates that a spectrally intense MS does not imply a longer total sequence duration, consistent with duration and spectral amplitude being governed by partially independent source and path effects. Accordingly, given its non-redundant information but limited efficiency in predicting component demand (Figs. 9 and 12), DS_{5-95} is best regarded as a secondary, complementary IM—most useful within vector-valued formulations—rather than as a stand-alone predictor for the components considered, in line with previous findings on ground-motion duration (Hancock and Bommer 2006; Chandramohan et al. 2016).

3.3 Engineering demand parameters and inter-component correlation

The seismic response of the bridge structure was quantified using a set of component-specific EDPs that reflect the dominant damage mechanisms and the progressive accumulation

of inelastic demand throughout MS and MS–AS sequences. The selection of EDPs was based on the mechanical behavior and vulnerability characteristics of each structural component, ensuring relevance for probabilistic fragility assessment.

For EB, the adopted EDP corresponds to the peak relative displacement between the deck and the substructure ($Disp^{EB}$), measured in both the longitudinal and transverse directions.

For BF, the EDP is defined as the maximum relative displacement between the backwall and the superstructure ($Disp^{BF}$). For COL, a cumulative DI was used to characterize nonlinear degradation processes. The adopted index follows the Park–Ang formulation (Park and Ang 1985), incorporating both peak drift and hysteretic energy dissipation, thereby accounting for strength deterioration and energy-based damage accumulation. The DI is computed as:

$$\mathrm{DI} = \frac{\Delta_{max}}{\Delta_u} + \tau \cdot \frac{E_{hyst}}{F_y \cdot \Delta_u} \tag{13}$$

where Δ_{max} is the maximum drift, Δ_u is the ultimate drift capacity, E_{hyst} is the cumulative hysteretic energy, F_y is the yield force, and $\tau = 0.15$ is an adopted energy-weighting coefficient. The value of τ is adopted as a modeling assumption rather than calibrated for the Águila Norte Bridge, following values recommended in the literature for RC columns with moderate-to-good transverse detailing (Park and Ang 1985; Cosenza et al. 1993; Teran-Gilmore and Jirsa 2005; Zhou et al. 2020). Since τ scales only the hysteretic-energy term and is applied consistently across all analyses, it affects the absolute damage-index values but not the relative comparison of candidate IMs. A component-specific calibration of τ is left for future work. The energy term was obtained from the integration of force–displacement hysteresis curves at the base of each COL using outputs from NLTHA.

Figure 6 shows the three-dimensional relationship between column DI and the displacement responses of EB and BF under MS-only (a) and MS–AS (b) loading. A strong positive correlation is observed among displacement-based EDPs. For MS-only sequences, regression surfaces achieve $R^2 = 0.87$, with Spearman and Pearson coefficients of 0.94 and 0.93, respectively. Under MS–AS, correlations remain high ($R^2 = 0.85$, Spearman = 0.92, Pearson = 0.92), though dispersion slightly increases due to cumulative AS effects.

To complement displacement-based metrics, a normalized performance variable $Y_{LS}^{(n)}$ was implemented (Ebrahimian et al. 2014; Jalayer and Ebrahimian 2017):

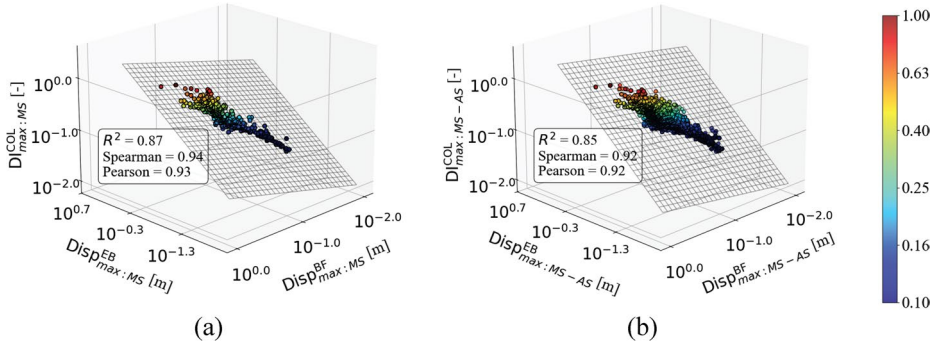

Fig. 6 Relationship between EDPs for: (**a**) MS; (**b**) MS–AS

$$Y_{LS}^{(n)} = \frac{D_{\max}^{(n)} - D_r^{(n-1)}}{C_{LS} - D_r^{(n-1)}} \tag{14}$$

where $D_{\max}^{(n)}$ is the demand from event n, $Dr^{(n-1)}$ the residual demand from previous events, and C_{LS} the capacity at the defined limit state (LS). Values close to 1 indicate that the current event alone exhausts the residual capacity, while values > 1 imply exceedance of the LS. This parameter captures cumulative degradation and identifies sequences where ASs trigger damage-state transitions.

The selected EDPs exhibit strong mutual correlation, largely because they are displacement-based and reflect the dominant seismic response mechanisms of the bridge. EB displacements, BF deformations, column-level indices (e.g., Park–Ang DI) and the other parameters $Y_{LS}^{(n)}$ show consistent trends under both MS and MS–AS loading, with correlation coefficients typically above 0.85. This high dependency is expected for RC highway bridges with integral pier caps and seat-type BF, where cumulative seismic demands manifest coherently across multiple components.

3.4 Definition of damage state thresholds for EDPs

The evaluation of structural performance under MS–AS sequences required defining consistent EDPs and discrete DS thresholds for each component. Reference values for this bridge typology are drawn from previous studies (Williams and Sexsmith 1995; Pinto et al. 2024; Guajardo et al. 2024).

For EB, the EDP is the relative displacement across the device, with thresholds set at 0.029 m for *Slight* damage (DS1), 0.104 m for *Moderate* damage (DS2), 0.136 m for *Extensive* damage (DS3), and 0.187 m for *Complete* failure (DS4).

In the case of BF, damage was characterized based on the horizontal displacement of the abutment wall. Thresholds correspond to 0.138 m for DS1, 0.152 m for DS2, 0.187 m for DS3, and 0.278 m for DS4, capturing the progressive loss of confinement and the onset of functional impairment in the abutment zone.

For COLs, the Park–Ang DI was adopted as the representative EDP. Damage thresholds for COLs were defined as 0.25 for slight, 0.40 for moderate, 0.85 for extensive, and 1.0 for complete damage, in alignment with established benchmarks for seismic damage in RC members.

4 Results

4.1 IM–EDP statistical evaluation

Figure 7 presents log–log scatter plots between five representative IMs (HI, $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, $Sa(T_1)$, and PGV) and the $Disp^{EB}$. These IMs were selected as top performers from the broader set screened in the study. Panels (a) to (d) correspond to alternative definitions of the IM–EDP pairing within the MS–AS framework.

In the MS-only configuration (Fig. 7a), period-matched spectral accelerations achieve the tightest clouds, with $Sa(T_1)_{\mathrm{MS}}$ and $Sa_{avg}(T_1)_{\mathrm{MS}}$ showing the strongest linear trends. This indicates high efficiency when the MS governs the response. Using the maximum IM over

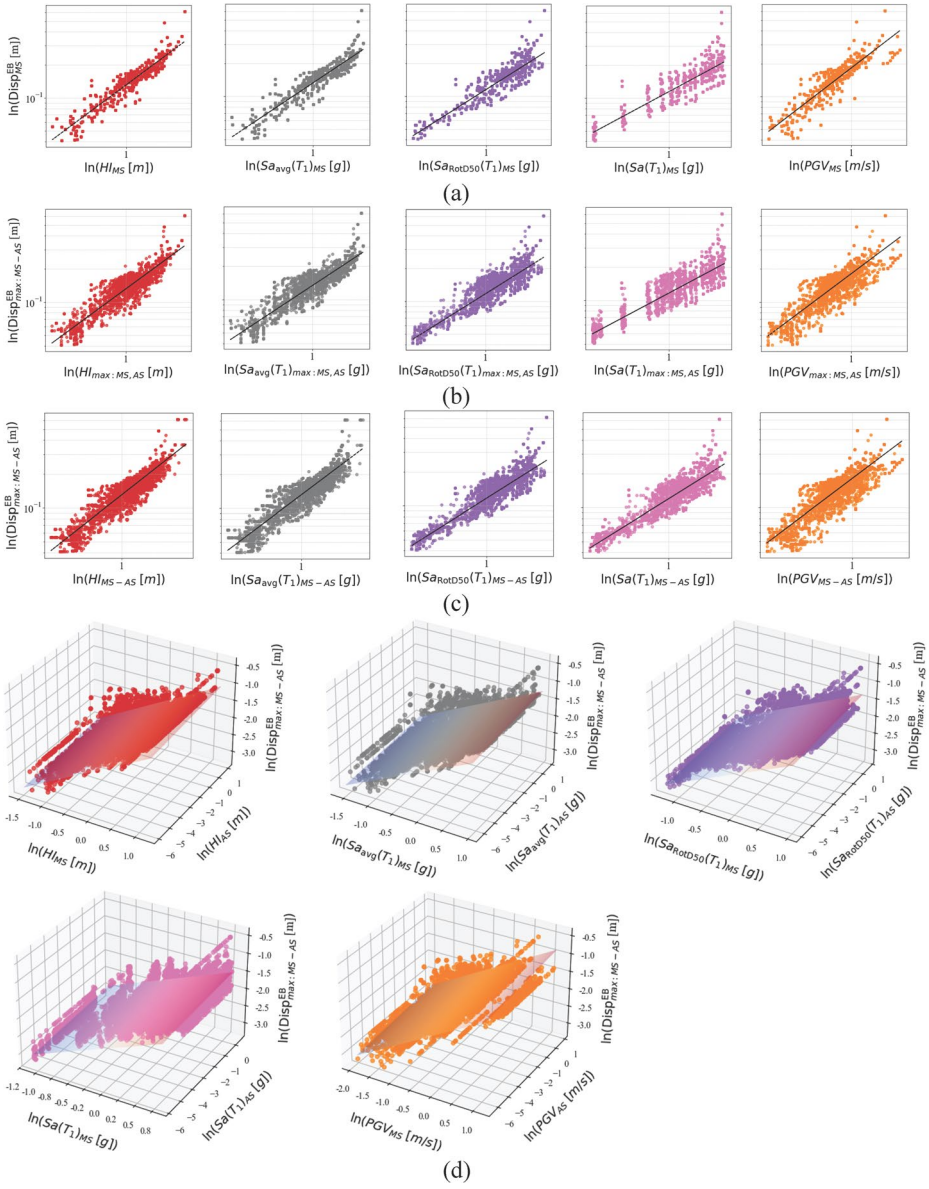

Fig. 7 EDP for EB vs. five representative IMs: (**a**) IM: MS & EDP: max (MS); (**b**) IM: max (MS, AS) & EDP: max (MS-AS); (**c**) IM: MS-AS & EDP: max (MS-AS); (**d**) IM: MS, AS & EDP: max (MS-AS)

the sequence (Fig. 7b) preserves the correlation with the cumulative peak $Disp^{EB}$ while slightly widening the scatter, reflecting variability introduced by ASs. Under this definition, velocity/energy-related measures (PGV, HI) improve relative to MS-only rankings, consistent with added sequence effects. Figure 7c focuses on IMs computed from the full MS–AS sequence (IM_{MS-AS}) and plotted against the sequence-maximum $Disp^{EB}$. It can be observed that fit quality decreases for all IMs, indicating that sequence-based predictors introduce

additional variability when AS contributions are included. Finally, in Fig. 7d dual-source IMs that combine MS and AS information yield the most balanced clouds across the five IMs. Improvements are most visible for $Sa_{avg}(T_1)$ and HI, supporting the benefit of MS–AS-informed predictors for cumulative response. Consequently, these identified 'optimal' traditional IMs are not merely comparative metrics but serve as the direct predictor variables for the PSDMs developed in this study. By integrating these specific efficiency-proven IMs, the proposed bridge risk model comprehensively accounts for the varying spectral and energy content of the full seismic sequence.

4.1.1 Efficiency

Figure 8 compares R^2 for 12 IMs across all EDPs (Sect. 3.3) under different MS–AS configurations. Grey traces represent individual EDPs, while the black lines denote the minimum, mean, and maximum envelopes of R^2 values, summarizing the overall dispersion of correlations among EDPs. The blue wedges highlight the three most efficient IMs, i.e., the three IMs achieving the highest median R^2. The near-parallel curves confirm the strong correlation among displacement-based EDPs. In the MS-only case (Fig. 8a), HI and $Sa_{avg}(T_1)$ consistently rank highest, followed by $Sa_{RotD50}(T_1)$. When combining MS and AS intensities using the maximum operator, max(MS, AS) (Fig. 8b), or the square root of the sum of squares, SRSS(MS, AS) (Fig. 8d), nearly identical rankings are obtained, while the full MS–AS sequence (Fig. 8c) shows $Sa(T_1)$ slightly overtaking HI. Overall, $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$,

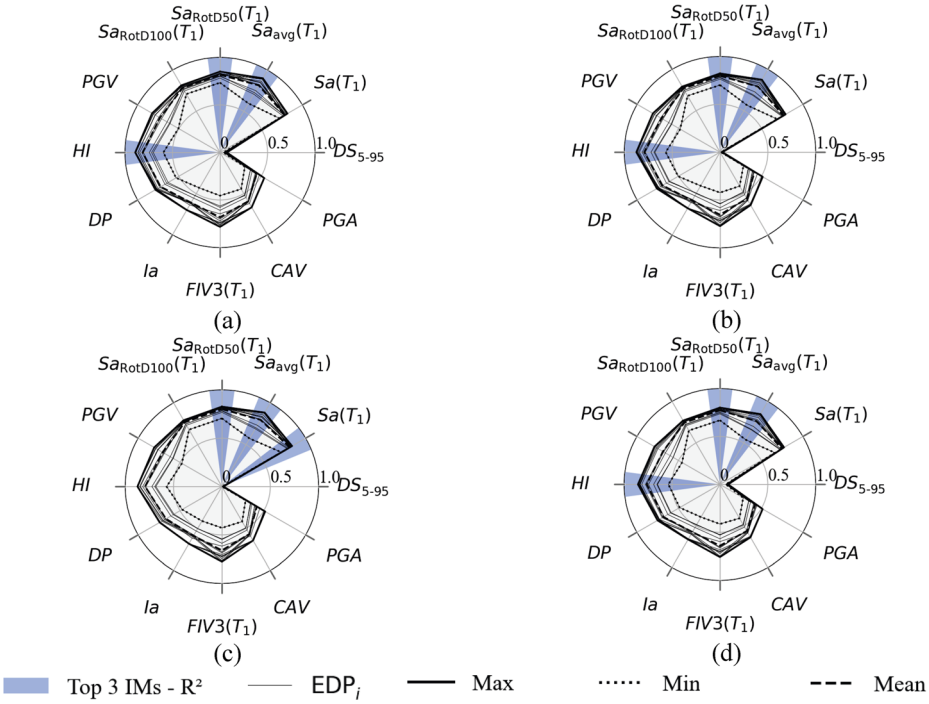

Fig. 8 Coefficient of determination (R^2): (**a**) IM: MS & EDP: max (MS); (**b**) IM: max (MS, AS) & EDP: max (MS-AS); (**c**) IM: MS-AS & EDP: max (MS-AS); (**d**) IM: SRSS (MS, AS) & EDP: max (MS-AS). Blue wedges highlight the three most efficient IMs (highest median R^2; higher is better)

HI, and $Sa(T_1)$ dominate across definitions, justifying the focus on MS-only and MS–AS sequence-based IMs in subsequent analyses.

Given the similarity in predictive trends between the MS-only IMs and those defined as max(MS, AS) or SRSS(MS, AS), subsequent analyses focus exclusively on the MS-only and MS–AS sequence-based IMs. This simplification avoids redundancy while preserving the relevant variability introduced by the AS contributions.

Figure 9 evaluates β of the IM–EDP relationships. In Fig. 9a, results show that $Sa_{RotD50}(T_1)$, $Sa_{avg}(T_1)$, and HI yield the lowest dispersion, indicating the highest efficiency according to the minimum-β criterion (lower β denotes higher efficiency), when the response is dominated by the MS. The top-3 IMs (in blue) maintain consistent performance across all EDPs, as indicated by the narrow β envelope. Figure 9b examines IMs over the full MS–AS sequence. While the leading IMs remain period-consistent—with $Sa(T_1)$ replacing $Sa_{RotD50}(T_1)$—the differences between IMs are less distinct, and the overall dispersion slightly increases due to AS-induced variability.

4.1.2 Practicality

Figure 10 presents the slope coefficient *b*, which measures the sensitivity of EDPs to IMs. A higher *b* value indicates greater practicality, reflecting a stronger and more stable scaling relationship between IMs and structural response. For MS-only (Fig. 10a), spectral acceleration-based IMs—$Sa(T_1)$, $Sa_{RotD50}(T_1)$, and $Sa_{RotD100}(T_1)$—show the steepest slopes, confirming their strong predictive capacity. In MS–AS sequences (Fig. 10b), slope variability increases slightly and mean values decrease, but the same period-consistent IMs remain dominant.

4.1.3 Proficiency

Figure 11 shows the proficiency index ζ, which balances dispersion (β) and slope (*b*). Lower ζ values denote higher predictive quality. For MS-only cases (Fig. 11a), $Sa(T_1)$, $Sa_{RotD50}(T_1)$, and HI achieve the best scores. Extending to MS–AS sequences (Fig. 11b), the

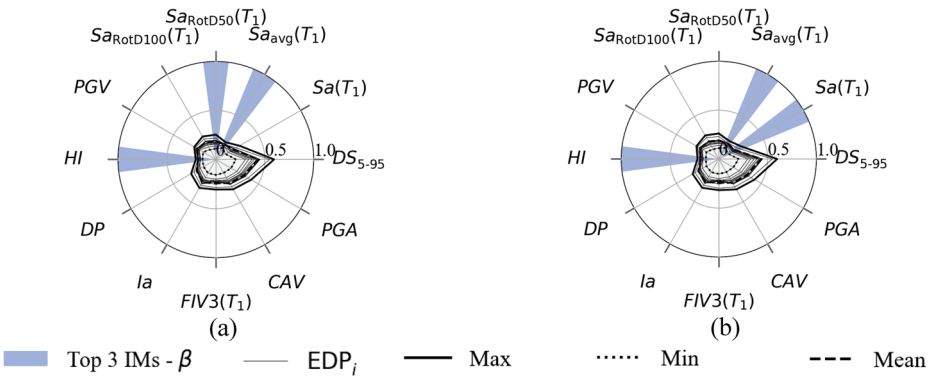

Fig. 9 Residual variability (β): (**a**) IM: MS & EDP: max (MS); (**b**) IM: MS-AS & EDP: max (MS-AS). Blue wedges highlight the three most efficient IMs (lowest median β; lower is better)

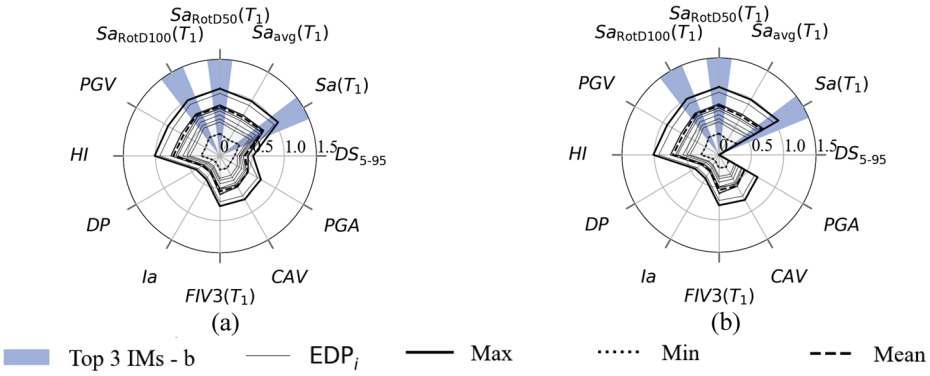

Fig. 10 Slope coefficient (*b*): (**a**) IM: MS & EDP: max (MS); (**b**) IM: MS-AS & EDP: max (MS-AS). Blue wedges highlight the three most practical IMs (highest median *b*; higher is better)

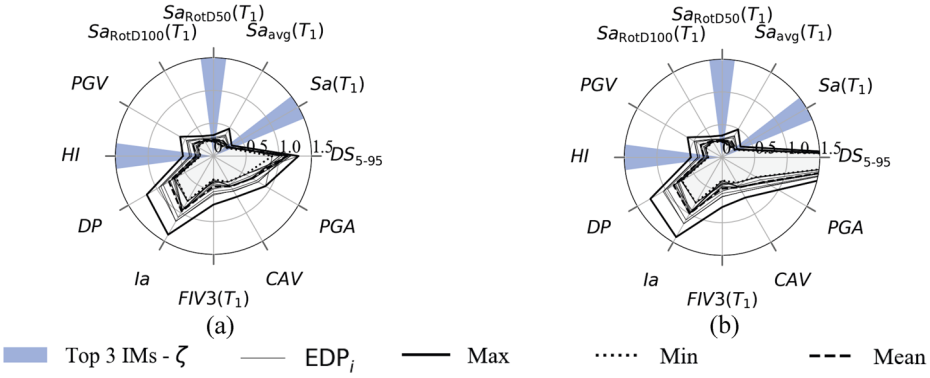

Fig. 11 Proficiency index (ζ): (**a**) IM: MS & EDP: max (MS); (**b**) IM: MS-AS & EDP: max (MS-AS). Blue wedges highlight the three best-performing IMs (lowest median ζ ; lower is better)

same IMs remain top-ranked, though ζ increases slightly and envelopes widen, reflecting the variability introduced by ASs.

4.1.4 Dispersion trends of IM–EDP models under MS–AS sequences

Figure 12 shows the evolution of dispersion ($\beta_{IM|EDP}$) across increasing levels of maximum relative $Disp^{EB}$. Subplots (a) to (c) correspond to IM–EDP relationships defined from MS-only values, maximum values between MS and AS, and full MS–AS sequences, respectively. Results indicate that dispersion tends to rise with demand—particularly beyond 0.20 m—highlighting reduced predictive efficiency of most IMs at higher damage levels. Parameters such as *Ia*, PGA, and DP consistently exhibit high β values, confirming their limited suitability for fragility modeling under severe response. In contrast, HI and spectral accelerations aligned with the fundamental period ($Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, $Sa(T_1)$) display stable and comparatively low dispersion across the entire EDP range. This reduction is further quantified through the PGA-normalized dispersion ratios reported in Appendix A (Table A.1 and Figure 19). For EB, HI

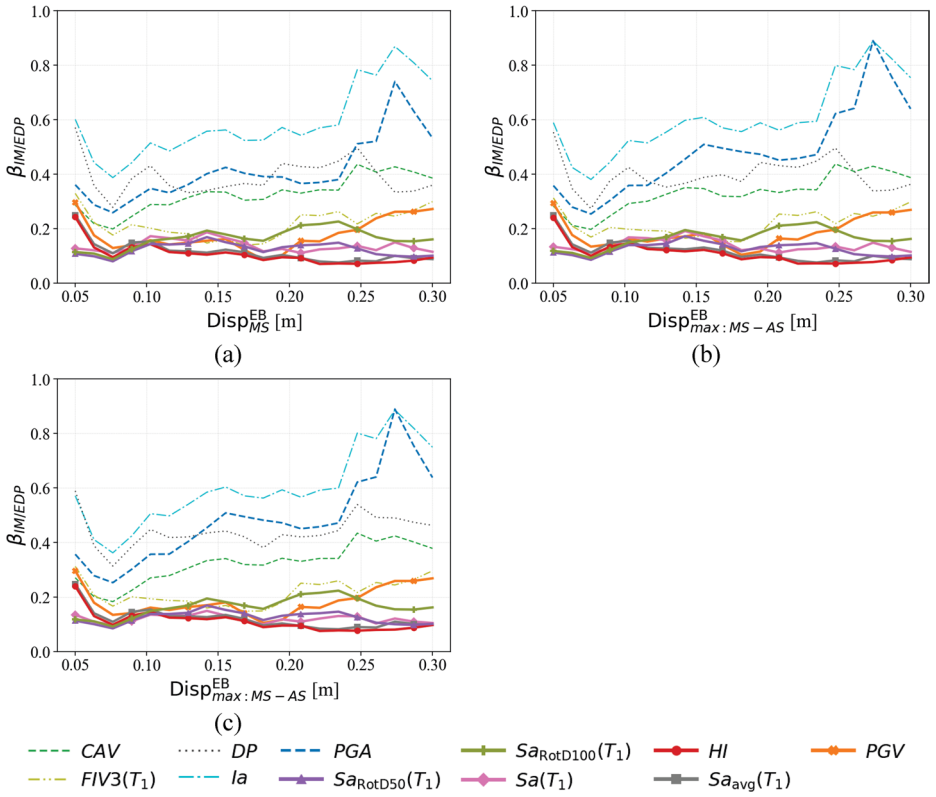

Fig. 12 Dispersion versus EDP ($Disp^{EB}$) for each IM investigated; (**a**) IM: MS & EDP: max (MS); (**b**) IM: max (MS, AS) & EDP: max (MS-AS); (**c**) IM: MS-AS & EDP: max (MS-AS)

exhibits mean β_{IM}/β_{PGA} values of 0.28, 0.26, and 0.26 for the MS-only, max(MS, AS), and full MS–AS definitions, respectively. These values correspond to average dispersion reductions of 72.3%, 73.9%, and 73.6% relative to PGA. Moreover, for 19 out of the 20 evaluated $Disp^{EB}$ thresholds, HI remains below $\beta_{IM}/\beta_{PGA}=0.5$, indicating that the reduction in threshold-wise dispersion is consistently greater than 50% over nearly the entire considered EB demand range. Including AS effects (Fig. 12b–c) modestly increases β, reflecting the variability introduced by sequential GMs. Nonetheless, spectral IMs tied to T_1 remain the most reliable predictors. This superior performance is attributed to the fact that the response of the analyzed bridge structure is strongly dominated by its fundamental mode of vibration. These IMs, therefore, effectively capture the primary dynamic behavior governing component demand and retain this robustness under cumulative seismic loading. It is acknowledged that this finding may not generalize to structures where higher-mode or torsional effects are more significant.

4.1.5 Exceedance probability surfaces for EB at DS2

Figure 13 shows the exceedance probability surfaces derived from the bivariate logistic regression model (Sect. 2.6) for EB at DS2, using alternative IM combinations for the MS and AS. Although all DS were analyzed (see Appendix B), a single representative DS is

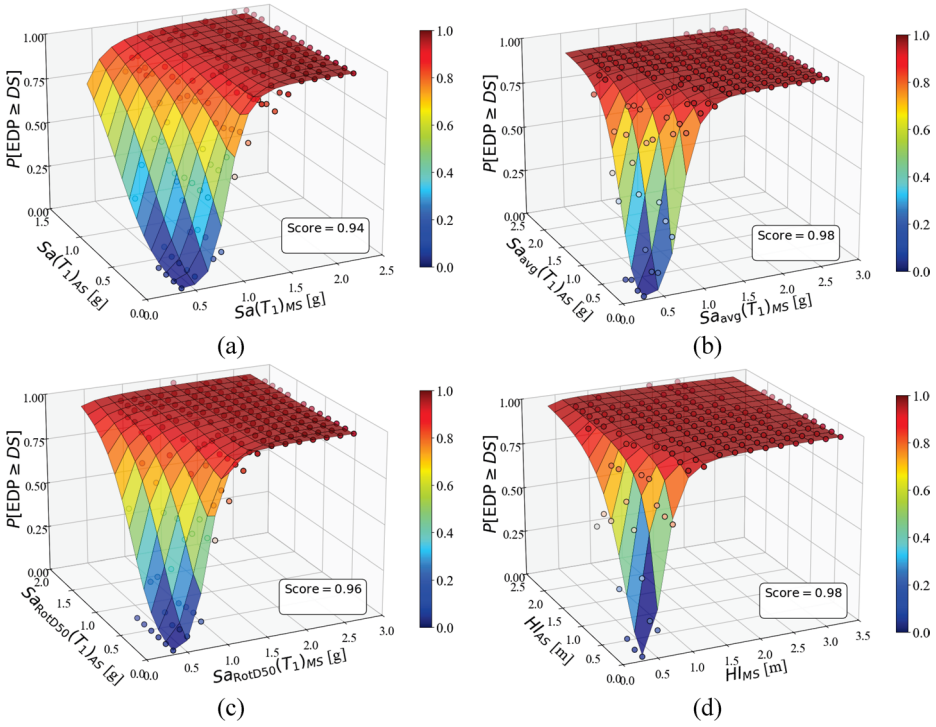

Fig. 13 Exceedance probability surface for EB at DS2 (δ_{max} = 0.104 m) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_{avg}(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

presented for each component—DS2 for EB, DS1 for BF, and DS1 for COL—to ensure comparability across components and focus on the most informative threshold in terms of onset of functional or structural impairment. When $Sa(T_1)$ is adopted for both events (Fig. 13a), the surface displays a steep nonlinear gradient with increasing exceedance probability as intensities grow. The gradient of the surface along the AS-intensity axis directly quantifies the cumulative effect: a steeper slope indicates that the AS intensity significantly increases failure probability for a fixed MS level. Although the classification accuracy is high (0.94), the transition zone remains moderately abrupt. $Sa_{avg}(T_1)$ (Fig. 13b) produces the smoothest transition and the highest predictive performance (0.98), effectively capturing the cumulative damage potential of sequential motions. Using $Sa_{RotD50}(T_1)$ (Fig. 13c) yields similarly robust predictions (0.96), with balanced gradients and consistent curvature. Finally, HI (Fig. 13d) achieves excellent accuracy (0.98) and produces a sharply defined surface, highlighting its capacity to predict DS2 exceedance in EB.

Overall, the comparison reveals that $Sa_{avg}(T_1)$ and HI consistently outperform other IMs in terms of predictive accuracy and surface coherence. While all IMs evaluated can capture the joint influence of MS and AS, those that integrate amplitude information over a range of periods (e.g., $Sa_{avg}(T_1)$) or incorporate energy-based metrics (e.g., HI) demonstrate superior proficiency in modeling exceedance under sequential loading. These insights support the use of such IMs in fragility surface generation and reinforce their suitability for performance-based seismic assessment frameworks that account for AS effects.

4.1.6 Exceedance probability surfaces for BF at DS1

Figure 14 presents the exceedance probability surfaces for BF at DS1. When $Sa(T_1)$ is employed for both MS and AS (Fig. 14a), the surface achieves moderate accuracy (0.91). The gradient rises more steeply along the MS axis, indicating that slight damage in BF is primarily governed by MS intensity, with limited AS influence at lower levels. Using $Sa_{avg}(T_1)$ (Fig. 14b) improves classification accuracy (0.94) and produces a smoother surface with a more symmetric gradient, evidencing the cumulative contribution of both events—consistent with BF's sensitivity to sequential loading. A comparable outcome is observed with $Sa_{RotD50}(T_1)$ (Fig. 14c), which attains the same accuracy (0.94) and exhibits uniform transitions across mid-range IM values, though with a less pronounced slope in the exceedance zone. When HI is adopted (Fig. 14d), the surface also achieves high accuracy (0.94) and delineates a sharper boundary between low- and high-probability regions, highlighting the robustness of energy-based IMs.

Among the IMs evaluated, HI consistently delivers the best classification performance, followed closely by $Sa_{avg}(T_1)$ and $Sa_{RotD50}(T_1)$. These results underscore the relevance of selecting IMs that reflect both amplitude and energy content for components susceptible to sequential loading effects.

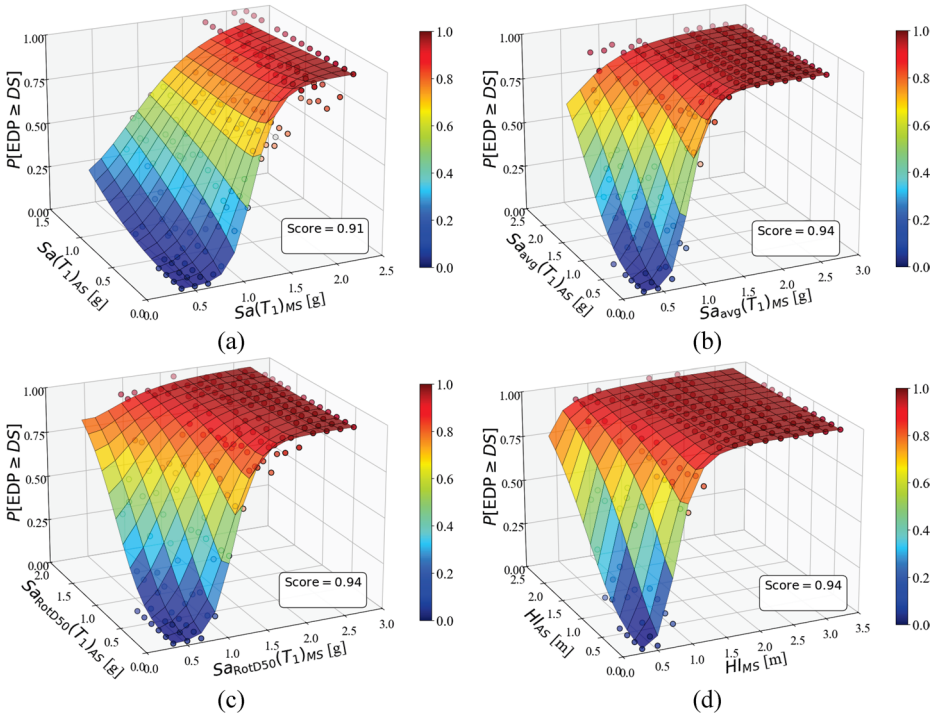

Fig. 14 Exceedance probability surface for BF at DS1 (δ_{max} = 0.1375 m) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_avg(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

4.1.7 Exceedance probability surfaces for COL at DS1

Figure 15 presents the exceedance probability surfaces for COLs at the onset of slight damage (DS1, DI = 0.25). When $Sa(T_1)$ is applied to both MS and AS (Fig. 15a), the surface achieves an accuracy of 0.92. However, the model fit is notably poor in the high-AS / low-MS intensity region. As highlighted in the figure, the surface in this zone systematically over-predicts the probability of exceedance, classifying many non-exceedance data points (blue circles) as having moderate probability.

$Sa_{avg}(T_1)$ (Fig. 15b) improves accuracy (0.95) and yields a significantly better model fit with smoother, more coherent gradients, reflecting its robustness in tracking cumulative COL demands. $Sa_{RotD50}(T_1)$ (Fig. 15c) performs similarly (0.93) and suffers from the same over-prediction deficiency at high AS intensities.

The clearest and most consistent separation is obtained with HI (Fig. 15d), which attains an accuracy of 0.94. The excellent fit to the data points across the entire domain and the sharply defined boundary underscore the effectiveness of velocity-based IMs in representing deformation potential and energy input for RC COLs under sequential shaking.

Overall, HI and $Sa_{avg}(T_1)$ emerge as the most proficient predictors of slight COL damage, combining high classification accuracy with a coherent and well-fitted surface definition. These IMs are particularly suitable for fragility surface generation and performance-based assessments that explicitly account for MS–AS effects.

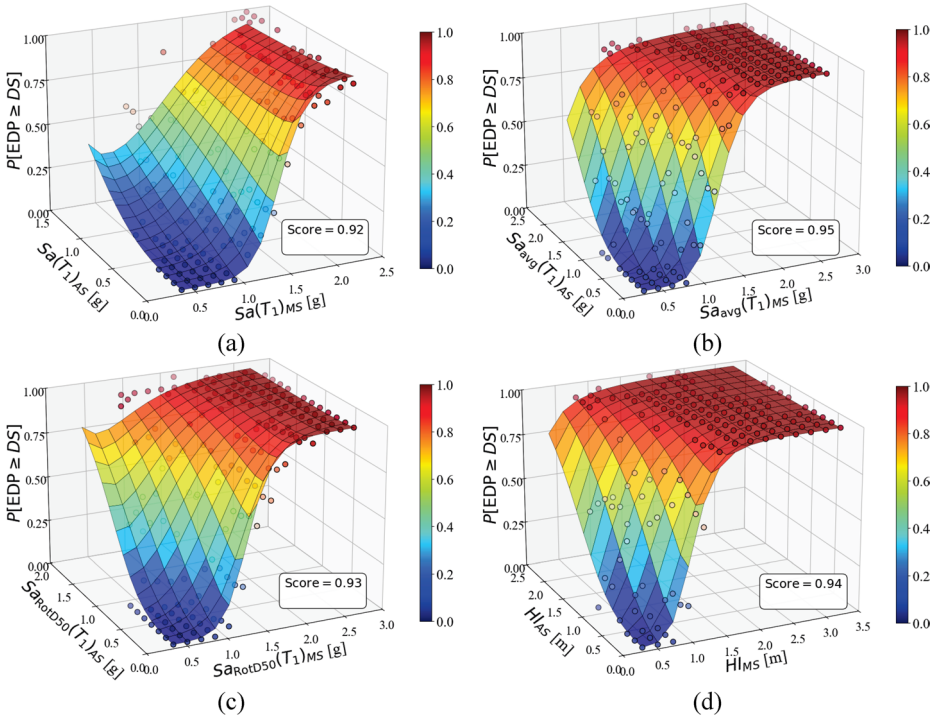

Fig. 15 Exceedance probability surface for COL at DS1 (DI = 0.25) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_{avg}(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

To verify whether the IM behavior observed in the representative exceedance-probability surfaces remains consistent at higher damage thresholds, additional surfaces were generated for the remaining component–damage-state combinations and are reported in Appendix B. These supplementary results show that the overall trends are preserved across damage states: $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, and HI generally provide more coherent probability surfaces than conventional peak or single-period IMs. As expected, the surfaces become more component- and threshold-dependent at higher damage states, reflecting the increasing influence of cumulative damage, residual deformation, and period elongation under MS–AS loading. Nevertheless, the supplementary results support the main conclusion that IMs incorporating spectral shape, orientation-independent response, and cumulative demand features are better suited for MS–AS fragility assessment.

4.1.8 Aftershock-induced exceedance-probability increment

To complement the bivariate exceedance-probability surfaces of Sects. 4.1.5–4.1.7, the AS contribution is quantified through an absolute exceedance-probability increment, ΔP, evaluated in the [IM_{MS}, IM_{AS}] space. For each fixed MS intensity, ΔP is defined relative to the lowest positive AS intensity available in the fitted grid, $IM_{\mathrm{AS,\,ref}}$, as given in Eq. (15):

$$\Delta P = P\left(IM_{\mathrm{MS}}, IM_{\mathrm{AS}}\right) - P\left(IM_{\mathrm{MS}}, IM_{\mathrm{AS,ref}}\right) \tag{15}$$

where P is the probability of exceeding the considered DS threshold. ΔP is thus an absolute increment, expressed in probability points.

Figure 16 shows representative ΔP maps for $Sa(T_1)$, $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, and HI, for EB–DS2, BF–DS1, and COL–DS1. The color field represents the AS-induced increment, while the black contours denote final exceedance-probability levels (P=0.25, 0.50, and 0.75), describing the locus of (IM_{MS}, IM_{AS}) pairs that yield each probability of exceeding the DS threshold. Across the representative cases, ΔP is predominantly positive and increases with IM_{AS}, confirming that ASs increase the probability of exceeding a given DS once the structure has already experienced a MS. This amplification is most pronounced for IMs that incorporate broader spectral, orientation-independent, or cumulative demand information: the maximum increment reaches 98.7 probability points for EB–DS2 with $Sa_{avg}(T_1)$, 83.0 for BF–DS1 with $Sa_{RotD50}(T_1)$, and 87.2 for COL–DS1 with $Sa_{avg}(T_1)$ (Table C.1).

The P=0.50 contours offer a complementary interpretation in terms of the MS intensity required to maintain a fixed exceedance-probability level. The associated reduction is defined in Eq. (16) as:

$$R_{IM_{\mathrm{MS}},p} = 100\left(1 - \frac{IM^{c}_{\mathrm{MS},p}}{IM^{c}_{\mathrm{MS},p,\mathrm{base}}}\right) \tag{16}$$

where $IM^{c}_{\mathrm{MS},p}$ is the MS intensity along the extracted $P=p$ contour and $IM^{c}_{\mathrm{MS},p,\mathrm{base}}$ is its value at the lowest-AS end of the same contour. For P=0.50, the maximum contour-based reductions in IM_{MS} reach 61.9%, 67.2%, and 76.4% for EB–DS2, BF–DS1, and COL–DS1, respectively; the same exceedance probability is thus attained with only 38%, 33%, and 24% of the baseline MS intensity as the relative AS contribution grows along the contour. The controlling IMs are $Sa_{avg}(T_1)$ for EB–DS2, $Sa_{\mathrm{RotD50}}(T_1)$ for BF–DS1, and $Sa_{avg}(T_1)$ for COL–DS1. The complete variation of $R_{IM_{\mathrm{MS}},p}$ with $IM^{c}_{\mathrm{AS,rel}}$ is provided in Appendix C

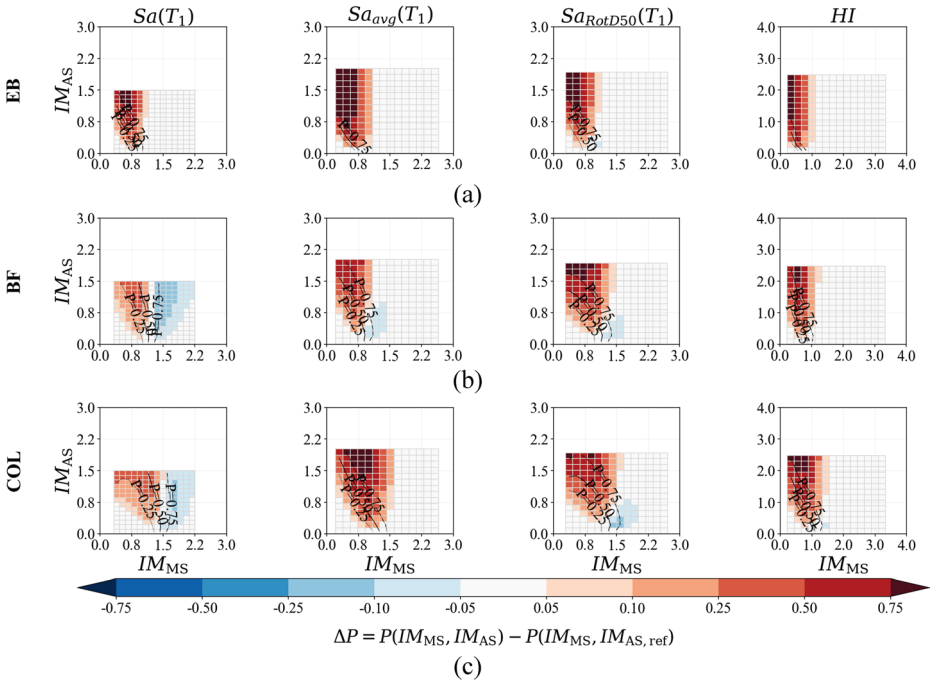

Fig. 16 Heatmaps of the aftershock-induced exceedance-probability increment, ΔP, for the four intensity measures $Sa(T_1)$, $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, and HI. Results are shown for: (**a**) EB–DS2 (δ_{max} = 0.104 m); (**b**) BF–DS1 (δ_{max} = 0.1375 m); and (**c**) COL–DS1 (DI = 0.25). Black contours indicate P = 0.25, 0.50, and 0.75. Complete results for all components and damage states are reported in Appendix C

for all components, DSs, and candidate IMs, where $IM^{c}_{AS,rel}$ denotes the normalized AS intensity along each extracted contour, ranging from the lowest-AS end to the highest-AS end of the contour.

Both diagnostics—the positive ΔP fields and the contour-based reductions in IM_{MS}—admit a consistent mechanical interpretation: the AS acts on a structure that may already be damaged by the MS, so that, at a fixed IM_{MS}, increasing IM_{AS} can raise the probability of exceeding a displacement- or damage-index threshold. This provides a direct probabilistic visualization of the non-conservatism of MS-only fragility models.

Some BF and COL maps exhibit isolated negative regions near the high-IM_{AS}/low-IM_{MS} corner. These should not be interpreted as ASs reducing damage potential; rather, they reflect local non-monotonicity of the fitted bivariate logistic surface in low-density regions of the (IM_{MS}, IM_{AS}) space, where predictions are more sensitive to edge extrapolation and limited empirical support. The interpretation of Fig. 16 therefore focuses on the central, data-supported region and on the relative behavior of the IMs, not on isolated extrapolated corners. Combinations lacking sufficient empirical contrast for a meaningful logistic fit were omitted rather than artificially extrapolated. The complete set of ΔP surfaces and tabulated metrics, reported in Appendix C, confirms that the AS-induced increment persists across DS, though its magnitude becomes increasingly component- and threshold-dependent at higher damage levels. This extended analysis supports the conclusion that optimal IMs for MS–AS fragility

assessment should capture not only amplitude, but also spectral shape, directionality, and cumulative energy-related features.

4.1.9 Comparative classification scores of IMs under MS–AS sequences

Figure 17 summarizes the classification performance of candidate IMs in predicting EDP threshold exceedance across bridge components. Each curve represents the logistic model score using paired IMs from MS and AS events (Sect. 2.6). A general decline in scores is observed with increasing EDP thresholds, reflecting the added dispersion and nonlinear damage accumulation at higher intensity levels. For EB (Fig. 17a), HI, $Sa_{avg}(T_1)$, and $Sa_{RotD50}(T_1)$ consistently achieve scores above 0.95, retaining discriminative capability even for large deformations (> 0.20 m). Specifically for EB, which are displacement-sensitive, neglecting the AS contribution in the IM selection leads to lower efficiency scores. The MS–AS sequences induce residual displacements that cumulative metrics capture better than peak MS-only values. In BF (Fig. 17b), the decline is more gradual; velocity- and energy-sensitive IMs (HI, PGV) remain the most effective, while PGA performs poorly. This efficiency superiority of HI over PGA stems from its integral nature. In subduction zones, where ground motion duration and energy content are significant, HI better captures the cumulative damage potential that instantaneous peak values like PGA miss. Frequency-

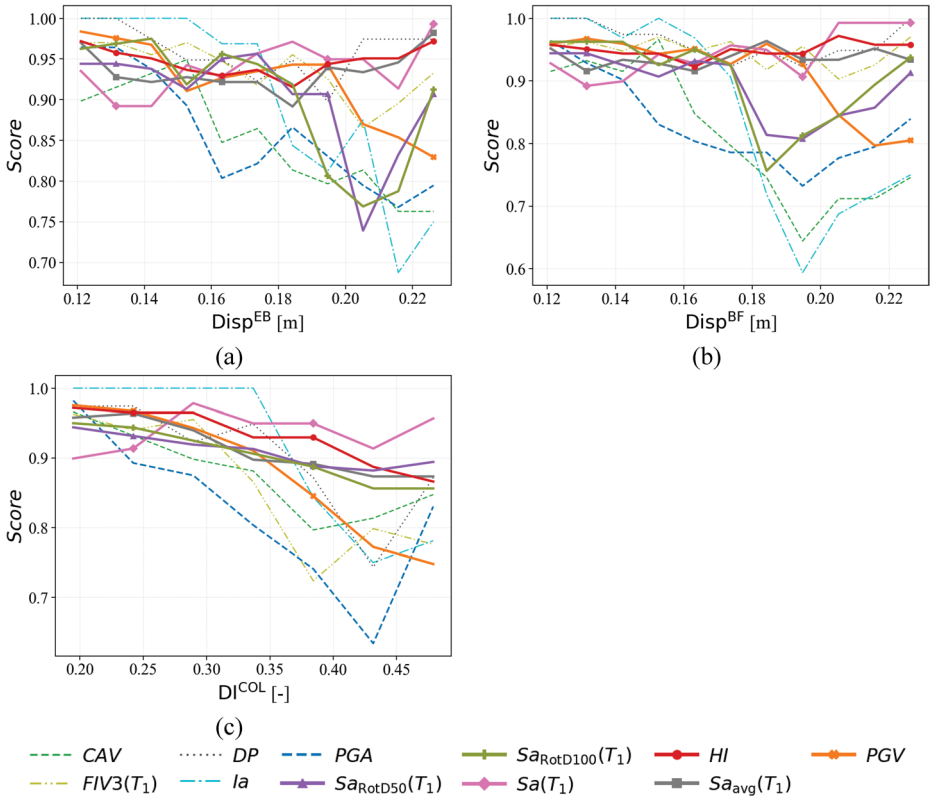

Fig. 17 Score versus EDP for each IM investigated: (**a**) EB; (**b**) BF; (**c**) COL

dependent metrics such as $Sa_{avg}(T_1)$ and $Sa_{RotD100}(T_1)$ also show robust performance. For RC COLs (Fig. 17c), $Sa_{avg}(T_1)$, HI, and $Sa(T_1)$ maintain the most stable classification across DI levels. As RC COLs degrade and soften during the sequence, their fundamental period elongates. $Sa_{avg}(T_1)$, by averaging spectral ordinates over a period range, implicitly accounts for this period elongation, thereby maintaining better correlation with damage than single-period $Sa(T_1)$. Whereas PGA and *Ia* deteriorate markedly with increasing damage thresholds. Overall, IMs that integrate amplitude, frequency content, and velocity-related features—particularly HI, $Sa_{avg}(T_1)$, and $Sa_{RotD50}(T_1)$—demonstrate superior stability and sufficiency across components. Regarding spectral acceleration metrics, $Sa_{RotD50}(T_1)$ is highlighted alongside the geometric mean $Sa(T_1)$ to address the aleatory uncertainty associated with arbitrary sensor orientation. By representing the median response across all non-redundant rotation angles, $Sa_{RotD50}(T_1)$ offers a theoretically more robust definition for components with direction-independent capacities, often enhancing sufficiency even where efficiency scores are comparable. In contrast, traditional peak-based measures such as PGA exhibit limited capability, especially under high-damage MS–AS conditions.

4.2 Sufficiency of IMs with respect to *M* and *R*

Figure 18 shows residual trends of EDPs against *M* and *R*, providing an assessment of IM sufficiency under MS-only and MS–AS conditions. Sufficiency is achieved when residuals display no systematic dependence on these seismological parameters, indicating that the IM fully captures source and path effects. For *M*, $FIV3(T_1)$ is the only IM that consistently satisfies sufficiency, particularly for column DI. In contrast, amplitude-based measures such as PGA, $Sa(T_1)$, and $Sa_{avg}(T_1)$ exhibit residual dependence on *M*, reflecting incomplete scaling with event size. For rupture distance, $Sa_{RotD100}(T_1)$ demonstrates strong sufficiency, with residuals remaining flat across the full *R* range, including near-field records. Other IMs, especially PGA, show moderate sensitivity, most evident for displacement-based responses, underscoring their limited capacity to account for attenuation effects. For columns, this distinction is particularly relevant: although $Sa_{RotD50}(T_1)$ attains slightly higher efficiency (lower β and higher R^2) and proficiency (lower ζ), $Sa_{RotD100}(T_1)$ is the only spectral IM achieving sufficiency with respect to rupture distance ($p=0.75$ and 0.24 for MS and MS–AS, respectively; Table 2), with non-significant residual dependence on *R*. Physically, because $Sa_{RotD100}(T_1)$ represents the maximum response over all orientations, it better reflects the orientation-critical drift and hysteretic-energy demand governing column damage, thereby absorbing the residual distance dependence that the median-based $Sa_{RotD50}(T_1)$ does not fully capture.

4.3 Assessing and identifying the optimal IMs

Table 2 summarizes the regression metrics linking seismic IMs with EDPs, including coefficient of R^2, *b*, β, ζ, and the *p*-values/slopes of residuals with respect to *M* and *R*. This multi-metric evaluation provides a rigorous basis for comparing IM performance across components and seismic scenarios. The results show that HI and $Sa_{avg}(T_1)$ consistently deliver the highest R^2 values—up to 0.84–0.89 for EB, BF, and COL—combined with moderate slopes ($b=0.72$–1.01), low dispersions ($\beta=0.16$–0.22), and proficiency indices (ζ) below 0.2–0.25. These trends confirm their strong predictive capacity and compact variability, making them the most proficient indicators of displacement demand in EB and BF,

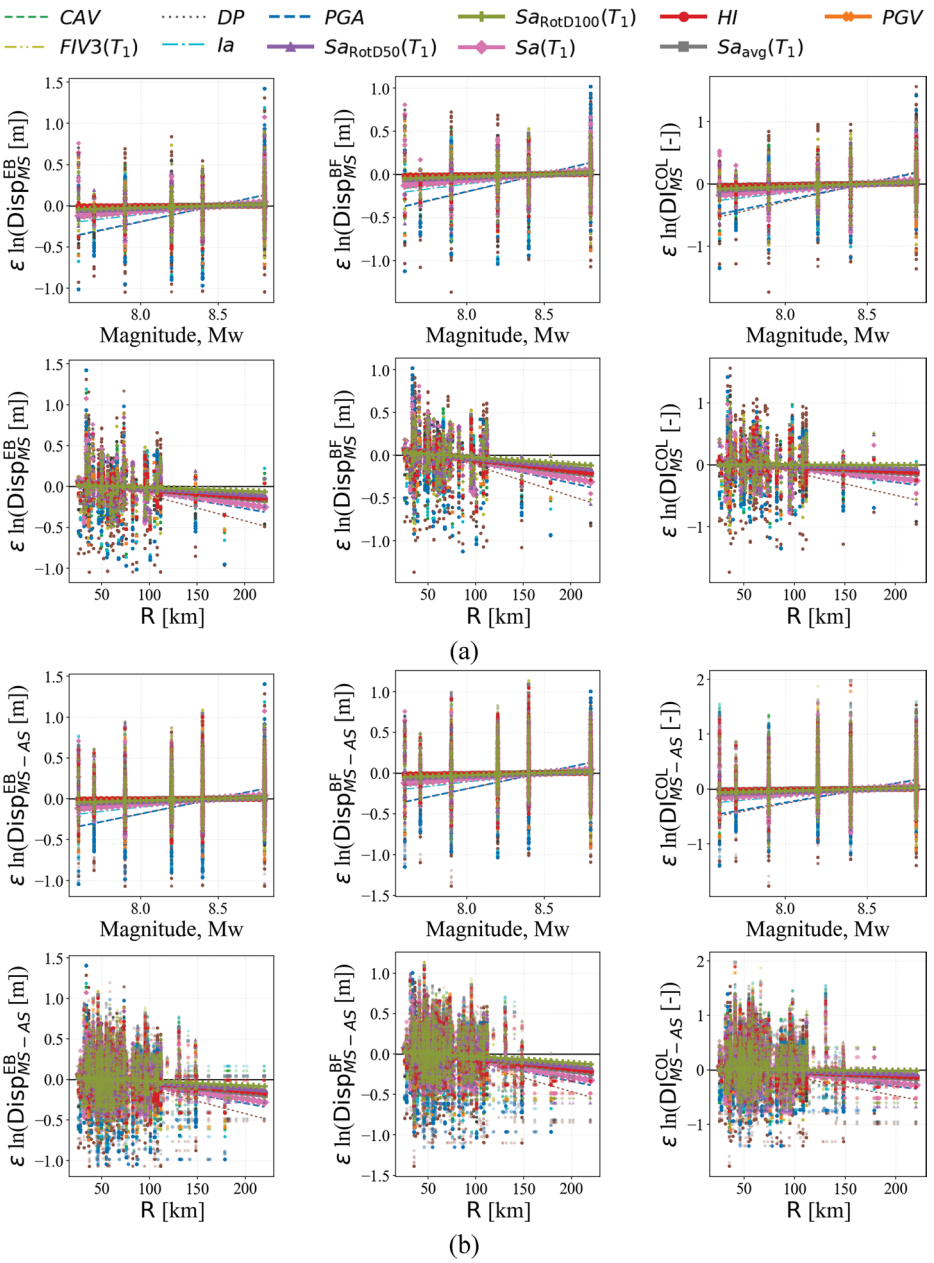

Fig. 18 Sufficiency with respect to magnitude and rupture distance of GMs for different EDPs; (**a**) MS; (**b**) MS-AS

while maintaining robustness for columns. $Sa_{RotD100}(T_1)$, though slightly lower in R^2 than HI and $Sa_{avg}(T_1)$, exhibits consistently low dispersion and ζ values, particularly under MS–AS sequences, highlighting its suitability for drift-sensitive components.

Table 2 Optimal IMs and statistical parameters including IM–EDP regression metrics and residual sufficiency with respect to *M* and *R*

Case	Component	IM	R^2	*b*	β	ζ	*p*-value		slope	
							M	*R*	*M*	*R*
MS	EB	**HI**	0.89	0.76	0.15	0.20	0.03	0.00	0.01	0.00
		$\boldsymbol{Sa_{avg}(T_1)}$	0.87	0.72	0.17	0.23	0.00	0.00	0.02	0.00
		$Sa_{RotD50}(T_1)$	0.82	0.78	0.19	0.25	0.00	0.00	0.06	0.00
		PGV	0.81	0.69	0.20	0.29	0.00	0.00	0.07	0.00
		$\boldsymbol{Sa_{RotD100}(T_1)}$	0.79	0.74	0.21	0.28	0.00	0.00	0.05	0.00
		$Sa(T_1)$	0.78	0.77	0.21	0.28	0.00	0.00	0.14	0.00
	BF	**HI**	0.86	0.77	0.17	0.22	0.00	0.00	0.02	0.00
		$\boldsymbol{Sa_{avg}(T_1)}$	0.85	0.73	0.18	0.25	0.00	0.00	0.03	0.00
		$Sa_{RotD50}(T_1)$	0.82	0.79	0.20	0.25	0.00	0.00	0.07	0.00
		PGV	0.79	0.70	0.21	0.30	0.00	0.00	0.08	0.00
		$\boldsymbol{Sa_{RotD100}(T_1)}$	0.79	0.76	0.22	0.29	0.00	0.00	0.06	0.00
		$Sa(T_1)$	0.78	0.80	0.22	0.28	0.00	0.00	0.15	0.00
	COL	$\boldsymbol{Sa_{avg}(T_1)}$	0.89	0.98	0.20	0.21	0.00	0.00	0.04	0.00
		HI	0.89	1.01	0.21	0.20	0.00	0.00	0.04	0.00
		$Sa_{RotD50}(T_1)$	0.83	1.05	0.25	0.24	0.00	0.00	0.10	0.00
		PGV	0.82	0.93	0.25	0.27	0.00	0.00	0.11	0.00
		$\boldsymbol{Sa_{RotD100}(T_1)}$	0.80	1.00	0.27	0.27	0.00	0.75	0.09	0.00
		$Sa(T_1)$	0.79	1.05	0.28	0.27	0.00	0.00	0.21	0.00
MS-AS	EB	**HI**	0.87	0.75	0.16	0.21	0.05	0.00	0.01	0.00
		$\boldsymbol{Sa_{avg}(T_1)}$	0.86	0.72	0.17	0.23	0.00	0.00	0.02	0.00
		$Sa(T_1)$	0.83	0.79	0.19	0.24	0.00	0.00	0.13	0.00
		$Sa_{RotD50}(T_1)$	0.82	0.78	0.19	0.24	0.00	0.00	0.06	0.00
		PGV	0.79	0.67	0.20	0.30	0.00	0.00	0.06	0.00
		$\boldsymbol{Sa_{RotD100}(T_1)}$	0.78	0.73	0.21	0.28	0.00	0.00	0.05	0.00
	BF	**HI**	0.85	0.76	0.17	0.23	0.00	0.00	0.02	0.00
		$\boldsymbol{Sa_{avg}(T_1)}$	0.84	0.73	0.18	0.25	0.00	0.00	0.03	0.00
		$Sa(T_1)$	0.83	0.81	0.19	0.23	0.00	0.00	0.14	0.00
		$Sa_{RotD50}(T_1)$	0.82	0.79	0.19	0.24	0.00	0.00	0.06	0.00
		$\boldsymbol{Sa_{RotD100}(T_1)}$	0.78	0.75	0.21	0.28	0.00	0.00	0.05	0.00
		PGV	0.77	0.68	0.22	0.32	0.00	0.00	0.08	0.00
	COL	$\boldsymbol{Sa_{avg}(T_1)}$	0.88	0.98	0.21	0.21	0.00	0.00	0.03	0.00
		HI	0.87	1.01	0.22	0.21	0.00	0.00	0.03	0.00
		$Sa(T_1)$	0.82	1.06	0.26	0.24	0.00	0.00	0.19	0.00
		$Sa_{RotD50}(T_1)$	0.82	1.05	0.26	0.24	0.00	0.00	0.09	0.00
		PGV	0.81	0.92	0.26	0.28	0.00	0.00	0.09	0.00
		$\boldsymbol{Sa_{RotD100}(T_1)}$	0.79	1.00	0.27	0.27	0.00	0.24	0.07	0.00

Note: Results are grouped into MS-only and MS–AS blocks. Within each component and case, IMs are ordered by decreasing (R^2). Bold/shaded IM names indicate the component-level recommended IMs selected from the combined efficiency, practicality, proficiency, and sufficiency assessment. Highlighted *p*-values indicate cases where the formal sufficiency criterion is satisfied (*p*-values > 0.05) with respect to *M* or *R*; highlighted residual slopes indicate cases with weak remaining dependence on *M*. *p*-values shown as 0.00 correspond to values rounded to two decimal places

The sufficiency analysis indicates that traditional IMs (PGA, PGV, $Sa(T_1)$) fail to meet independence criteria, showing significant residual dependence on M and R. In contrast, HI and $Sa_{avg}(T_1)$ achieve near-zero residual slopes and non-significant p-values in most cases, especially for EB and BF. $Sa_{RotD100}(T_1)$ further stands out as the most distance-sufficient IM, with residual slopes < 0.07 and p-values > 0.05, confirming resilience to path- and direction-dependent effects typical of subduction-related MS–AS sequences.

In summary, HI, $Sa_{avg}(T_1)$, and $Sa_{RotD100}(T_1)$ emerge as the optimal IMs for fragility analysis. HI and $Sa_{avg}(T_1)$ are recommended for displacement-sensitive components such as EB and BF, whereas $Sa_{RotD100}(T_1)$ is more effective for drift-sensitive measures like column DI, mainly because it provides the best distance-sufficiency for columns within the multi-metric ranking, rather than because it is uniformly superior to $Sa_{RotD50}(T_1)$ across all statistical metrics. These findings reinforce the need for multi-metric IM selection frameworks in subduction zones, where conventional IMs fail to capture the compounded demands of sequential earthquakes.

4.4 Limitations and future work

The component-level IM rankings reported in this study should be interpreted in relation to three factors: the structural configuration of the case-study bridge, the modeling assumptions adopted, and the in-sample nature of the statistical evaluation.

First, the rankings are conditional on the structural typology of the Águila Norte Bridge, which comprises simply-supported spans, elastomeric bearings, seat-type abutments, and reinforced-concrete columns. This configuration yields response mechanisms strongly influenced by relative deck–support displacements, bearing deformation, deck–abutment interaction, and cumulative MS–AS sequence effects, and explains the strong performance of broad-band and velocity-/energy-sensitive IMs such as HI and $Sa_{avg}(T_1)$ for displacement-sensitive components: the flexibility introduced by the elastomeric bearings shifts part of the response toward longer-period, velocity-sensitive demands, reducing the descriptive capacity of short-period peak-acceleration measures, while the seat-type supports and simply-supported spans render displacement accumulation and sequence-induced demand particularly relevant for bearings and abutment backfill. For bridges with different configurations — monolithic or integral-abutment systems, fixed-bearing or short-period structures, or alternative seismic isolation devices — the controlling demand mechanisms, and hence the optimal IMs, may differ.

Second, the rankings are based on a fixed-base structural model. Soil–structure interaction may lengthen the effective period, introduce additional damping, and redistribute demand among components, particularly under the long-duration subduction records characteristic of the Chilean environment. The expected direction of this effect is that IMs anchored to the fixed-base fundamental period, such as $Sa(T_1)$ and rotated spectral accelerations evaluated at T_1, may lose efficiency as the effective period shifts, whereas band-integrated measures such as $Sa_{avg}(T_1)$, evaluated over $0.2T_1$–$3T_1$, and HI, which integrates spectral velocity over a broad period range, are expected to be less sensitive to moderate period shifts. The proposed workflow can be directly extended to models incorporating soil springs, pile–soil interaction, or other SSI formulations.

Third, the statistical evaluation is in-sample: the PSDMs and the associated efficiency, practicality, proficiency, and sufficiency metrics were computed from the same NLTHA dataset. While consistent with common PSDM-based fragility applications, this approach does not constitute an independent predictive validation and may introduce some degree of

overfitting, particularly for proficiency, which depends directly on the fitted PSDM parameters. The reported rankings should therefore be viewed as in-sample comparative rankings for the analyzed bridge and ground-motion dataset.

Accordingly, the proposed evaluation framework is generalizable, but the component-specific IM recommendations should be re-evaluated for other bridge classes, support and foundation conditions, and against independent data. Future work should assess the stability of the selected IMs using cross-validation and out-of-sample strategies — including k-fold cross-validation, repeated train/test splits, leave-one-sequence-out validation, and testing against independent hazard-consistent MS–AS sequences — and should explicitly quantify SSI effects on the IM rankings, thereby distinguishing true predictive performance from dataset-specific fitting effects.

5 Conclusions

This study introduced a probabilistic, multi-metric framework to identify optimal seismic intensity measures (IMs) for the fragility assessment of reinforced-concrete bridge components subjected to mainshock–aftershock (MS–AS) sequences in subduction zones. The framework integrated advanced hazard characterization, stochastic AS modeling, hazard-consistent ground motion (GM) selection, nonlinear dynamic analyses, and a comprehensive evaluation of IMs through efficiency, practicality, proficiency, and sufficiency criteria. Its application to the Águila Norte Bridge in Chile provided critical insights into both methodological development and practical implications.

The results confirmed the essential role of ASs in fragility assessment. Neglecting ASs underestimated component-level damage-exceedance probabilities, particularly at moderate and extensive damage states. The bivariate ΔP maps and the exceedance-probability contour diagnostics ($P = 0.50$; Sect. 4.1.8) indicated that, in the most pronounced representative cases, the same exceedance level is attained with as little as 38%, 33%, and 24% of the baseline MS intensity for the elastomeric bearings (EB), abutment backfill (BF), and columns (COL), respectively, as the AS contribution increases along the contour. This finding demonstrates that MS-only fragility models are non-conservative in subduction environments, where ASs are both frequent and intense. This non-conservatism arises because traditional single-event IMs ignore the energy content of the AS and fail to account for pre-existing damage from the MS, thereby predicting a higher remaining capacity than physically exists. The performance of IMs was shown to be strongly component specific. For displacement-sensitive components such as EB and BF, velocity- and spectrum-averaged measures (e.g., Housner Intensity and $Sa_{avg}(T_1)$) provided the most reliable predictions, while column fragility was better captured by $Sa_{avg}(T_1)$ and rotated spectral accelerations than by conventional parameters such as PGA and $Sa(T_1)$. These traditional IMs exhibited poor proficiency and sufficiency under sequential loading, with persistent dependencies on magnitude and rupture distance, reflecting their inability to account for cumulative energy and duration effects.

The multi-metric evaluation framework proved particularly advantageous, as no single criterion alone was sufficient to establish the robustness of an IM. By jointly assessing efficiency, practicality, proficiency, and sufficiency, the framework enabled transparent ranking of candidate measures and reduced bias toward IMs that perform well individually. HI and $Sa_{avg}(T_1)$ emerged as consistently optimal across metrics, balancing predictive power, physical interpretability, and compatibility with hazard characterization. From a methodological perspective, the integration of AS-conditioned probabilistic seismic hazard assessment, Chile-calibrated ETAS-based AS scenarios, recurrent-neural-network GM prediction, and hazard-consistent record selection ensured physically realistic MS–AS input motions. This allowed fragility models and conditional exceedance surfaces to capture cumulative degradation and loss of residual capacity in a manner consistent with subduction-zone seismicity.

Practically, this study provides actionable guidance for component-level assessment, particularly for structures where the response is dominated by the first mode. For displacement-sensitive components (EB and BF), the findings strongly recommend HI or $Sa_{avg}(T_1)$ as optimal predictors. For flexure-dominated components (COL), $Sa_{avg}(T_1)$ provided the strongest overall predictive performance, while $Sa_{RotD100}(T_1)$ is retained as a complementary recommended IM owing to its superior sufficiency with respect to rupture distance, particularly under MS–AS loading. Finally, component-specificity is critical because stiff components (columns) and flexible isolators (bearings) respond to different ground motion features (acceleration vs. displacement), rendering a 'one-size-fits-all' IM ineffective for holistic resilience assessment.

Despite these contributions, the study's limitations must be acknowledged to frame future work. The findings are based on a fixed-base structural model, neglecting potential soil-structure interaction (SSI) effects. Furthermore, the joint-hazard framework was built in the $Sa(T_1)$ space, and the IM evaluation metrics were computed in-sample. Future research should explore the influence of SSI, adopt more sophisticated component models, and validate IM proficiency using out-of-sample datasets or alternative hazard-anchoring IMs.

This highlights a key distinction: while the specific findings (e.g., HI being optimal for EBs) may not be universally generalizable, the proposed framework is. It provides a validated, robust protocol for identifying reliable IMs at the component level. The framework is generalizable to other bridge typologies and seismic contexts, offering a pathway toward more realistic fragility modeling and more resilient infrastructure planning in regions exposed to megathrust earthquakes.

Appendix A. PGA-normalized threshold-wise dispersion

Table A.1 summarizes the PGA-normalized threshold-wise dispersion, and Figure 19 illustrates its variation with increasing EB displacement threshold.

The relative dispersion ratio β_{rel} is defined as β_{IM}/β_{PGA}, where βIM is the fitted threshold-wise dispersion associated with a given IM and β_{PGA} is the corresponding dispersion obtained using PGA at the same $Disp^{EB}$ threshold. Therefore, $\beta_{rel} < 1.0$ indicates lower dispersion than PGA, whereas $\beta_{rel} \leq 0.5$ indicates a reduction of at least 50% relative to PGA.

Table A.1 Summary of PGA-normalized threshold-wise dispersion for EB

Case	IM	Mean β_{rel}	Mean reduction vs. PGA (%)	Max reduction vs. PGA (%)
IM: MS & EDP: max (MS)	HI	0.28	72.31	89.66
	$Sa_{avg}(T_1)$	0.30	70.08	86.50
	$Sa_{RotD50}(T_1)$	0.32	68.12	86.56
	$Sa(T_1)$	0.35	65.15	79.82
	PGV	0.43	56.70	77.95
	PGA	1.00	0.00	0.00
IM: max (MS, AS) & EDP: max (MS-AS);	HI	0.26	73.90	91.37
	$Sa_{avg}(T_1)$	0.28	72.02	88.81
	$Sa_{RotD50}(T_1)$	0.29	71.45	88.76
	$Sa(T_1)$	0.31	68.59	83.37
	PGV	0.40	59.78	78.47
	PGA	1.00	0.00	0.00
IM: MS-AS & EDP: max (MS-AS)	HI	0.26	73.58	90.97
	$Sa_{avg}(T_1)$	0.28	71.58	87.69
	$Sa_{RotD50}(T_1)$	0.28	71.62	88.76
	$Sa(T_1)$	0.27	72.64	86.41
	PGV	0.40	59.59	78.22
	PGA	1.00	0.00	0.00

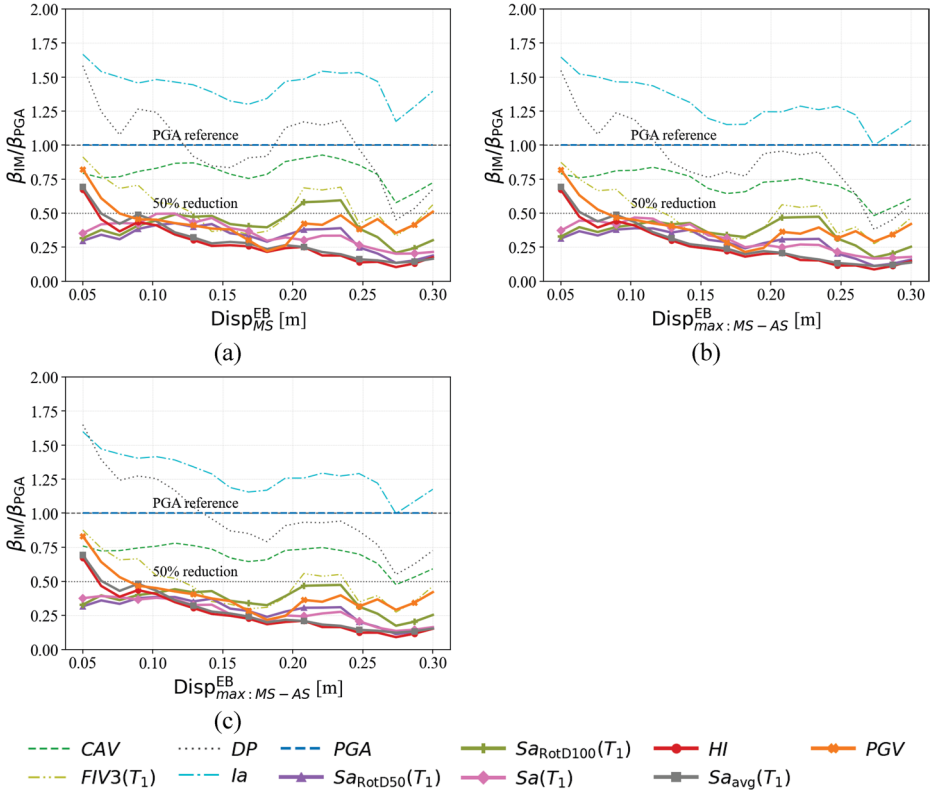

Fig. A.1 Relative threshold-wise dispersion versus EDP ($Disp^{EB}$) for each IM investigated, normalized with respect to PGA. The plotted quantity is $\beta_{rel} = \beta_{IM}/\beta_{PGA}$, where values below 1.0 indicate lower dispersion than PGA and values near or below 0.5 indicate an approximately 50% reduction. Panels correspond to: (**a**) IM: MS & EDP: max (MS); (**b**) IM: max (MS, AS) & EDP: max (MS-AS); (**c**) IM: MS-AS & EDP: max (MS-AS)

Appendix B. Supplementary exceedance-probability surfaces across damage states

Figures 20–28 present the supplementary exceedance-probability surfaces for the remaining component–damage-state combinations.

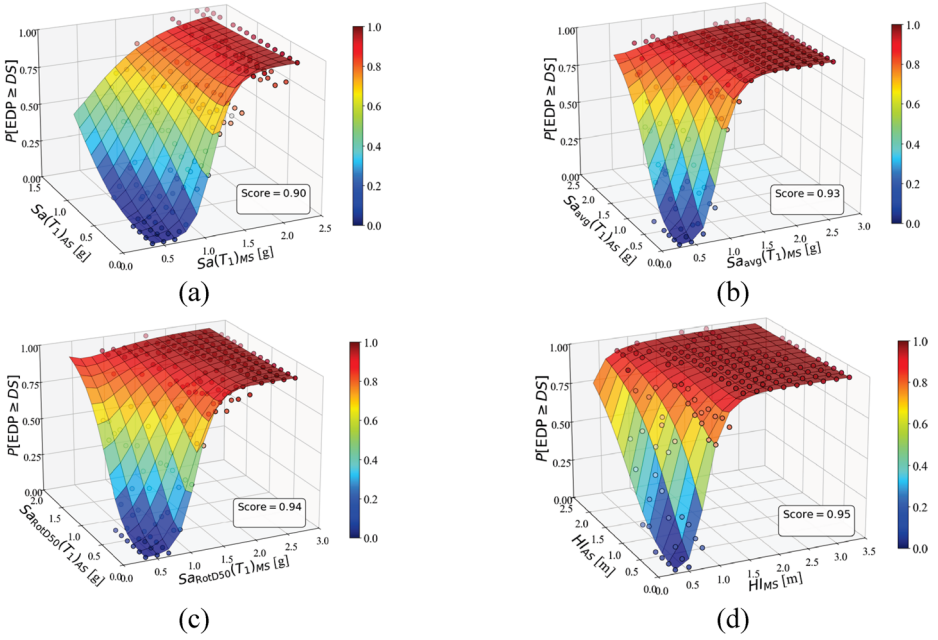

Fig. B.1 Exceedance probability surface for EB at DS3 (δ_{max} = 0.136 m) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_{avg}(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

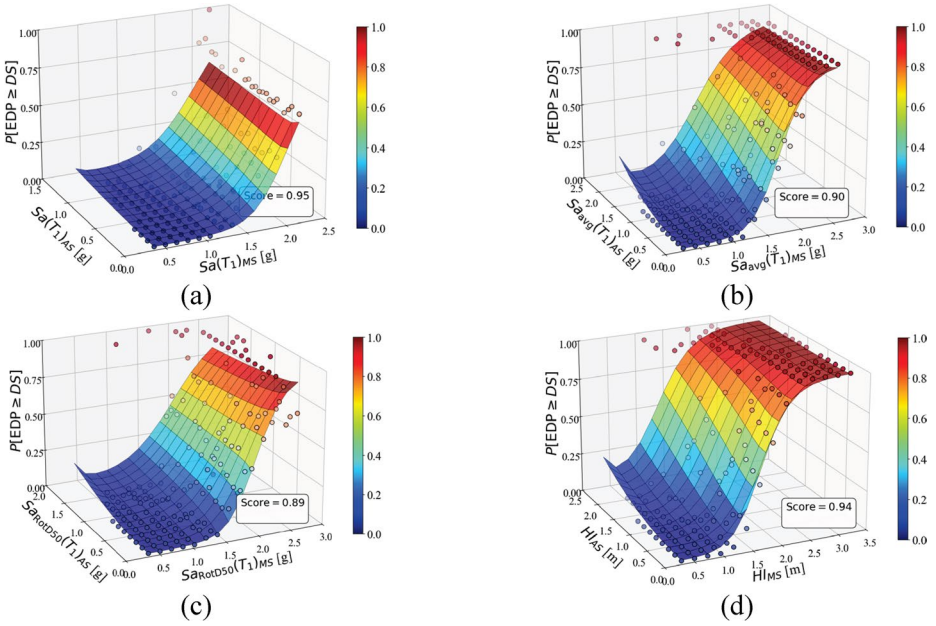

Fig. B.2 Exceedance probability surface for EB at DS4 (δ_{max} = 0.187 m) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_{avg}(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

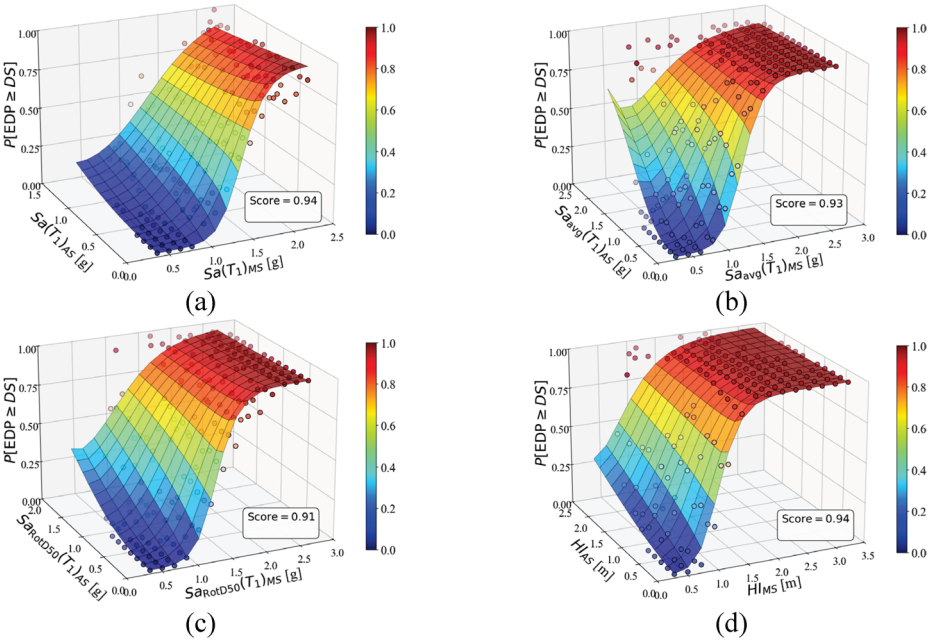

Fig. B.3 Exceedance probability surface for BF at DS2 (δ_{max} = 0.152 m) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_avg(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

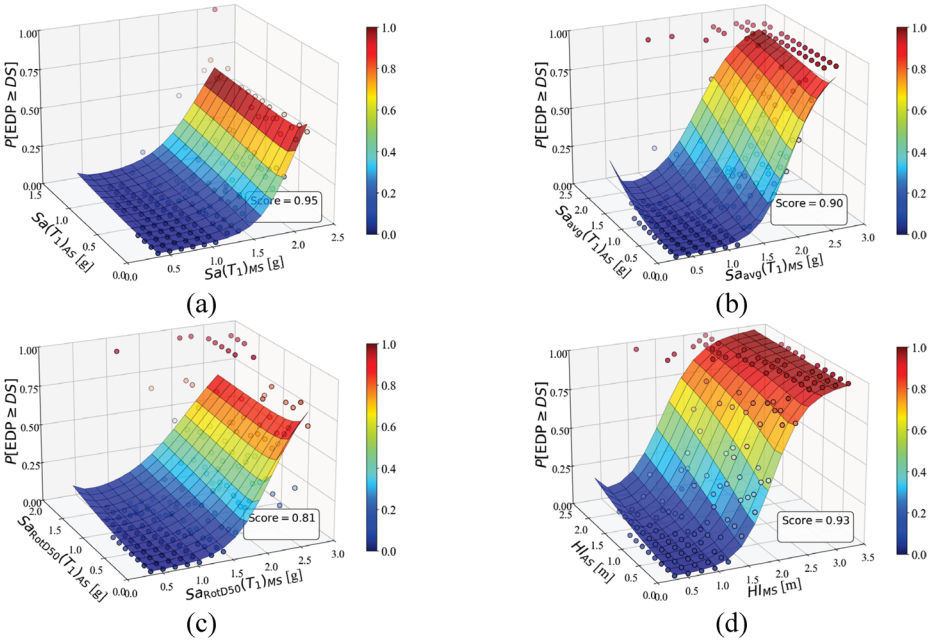

Fig. B.4 Exceedance probability surface for BF at DS3 (δ_{max} = 0.187 m) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_avg(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

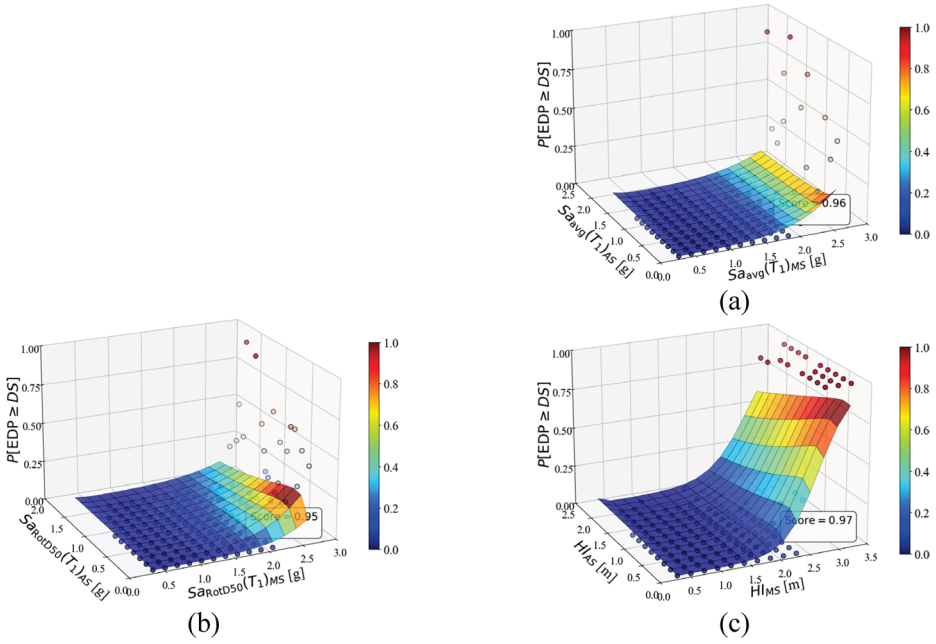

Fig. B.5 Exceedance probability surface for BF at DS4 (δ_{max} = 0.278 m) using different IM: (**a**) $Sa_avg(T_1)$; (**b**) $Sa_{RotD50}(T_1)$; (**c**) HI. $Sa(T_1)$ is omitted because no valid bivariate logistic surface was retained for this case

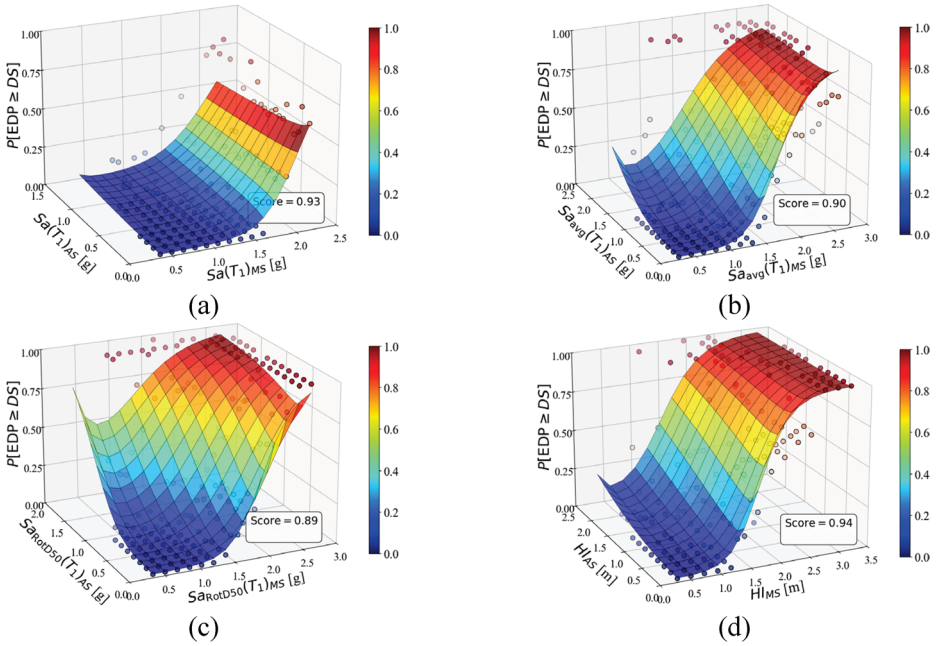

Fig. B.6 Exceedance probability surface for COL at DS2 (DI = 0.40) using different IM: (**a**) $Sa(T_1)$; (**b**) $Sa_{avg}(T_1)$; (**c**) $Sa_{RotD50}(T_1)$; (**d**) HI

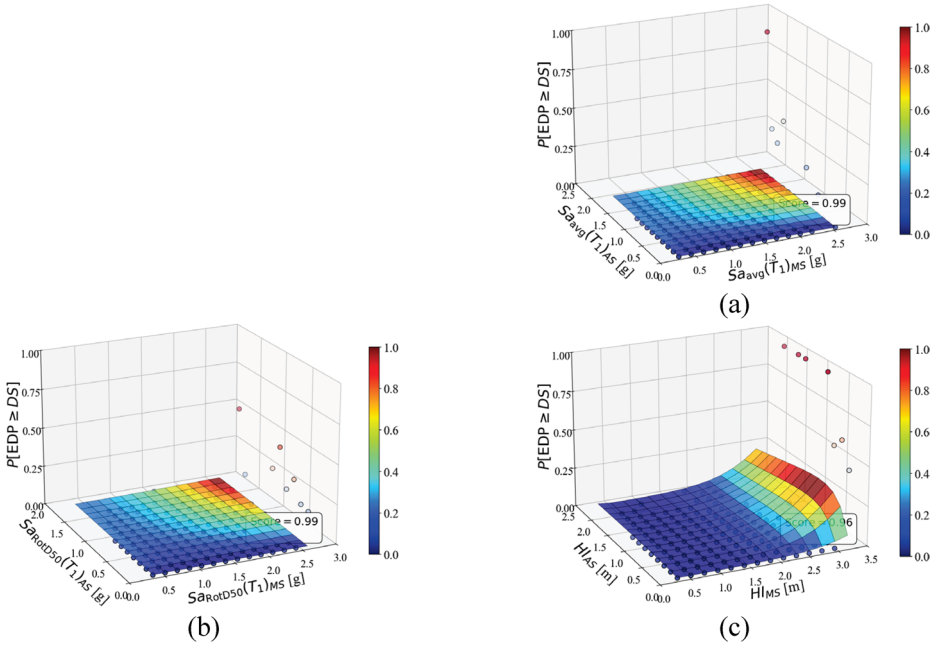

Fig. B.7 Exceedance probability surface for COL at DS3 (DI = 0.85) using different IM: (**a**) $Sa_{avg}(T_1)$; (**b**) $Sa_{RotD50}(T_1)$; (**c**) HI. $Sa(T_1)$ is omitted because no valid bivariate logistic surface was retained for this case

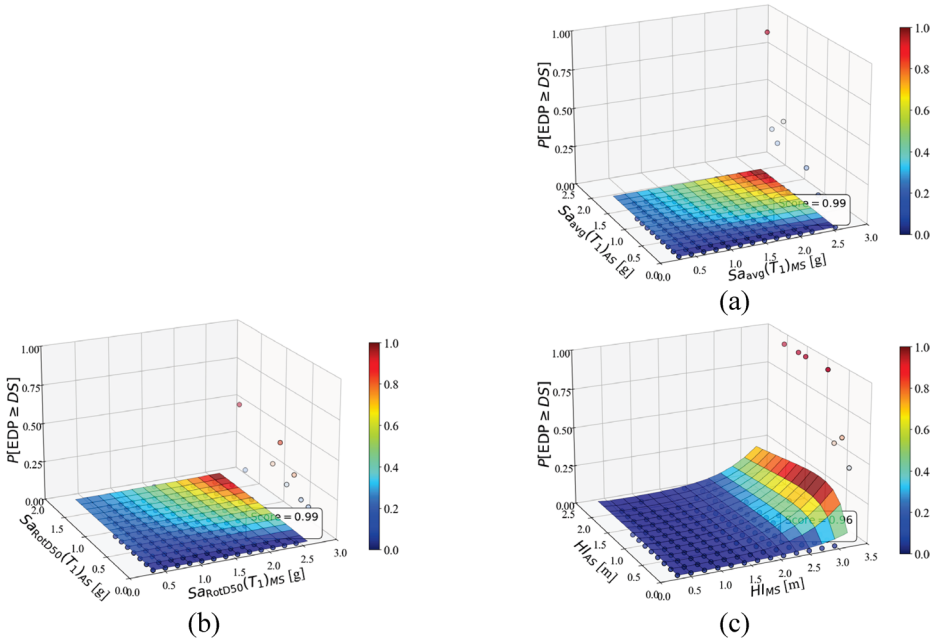

Fig. B.8 Exceedance probability surface for COL at DS4 (DI = 1.00) using different IM: (**a**) $Sa_{avg}(T_1)$; (**b**) $Sa_{RotD50}(T_1)$; (**c**) HI. $Sa(T_1)$ is omitted because no valid bivariate logistic surface was retained for this case

Appendix C. Supplementary Δ*P* surfaces for all components and damage states (DS)

This appendix provides the supplementary probability-space diagnostics supporting Sect. 4.1.8. Figures 29–30 extend the Δ*P* heatmaps to the remaining valid component–DS combinations.

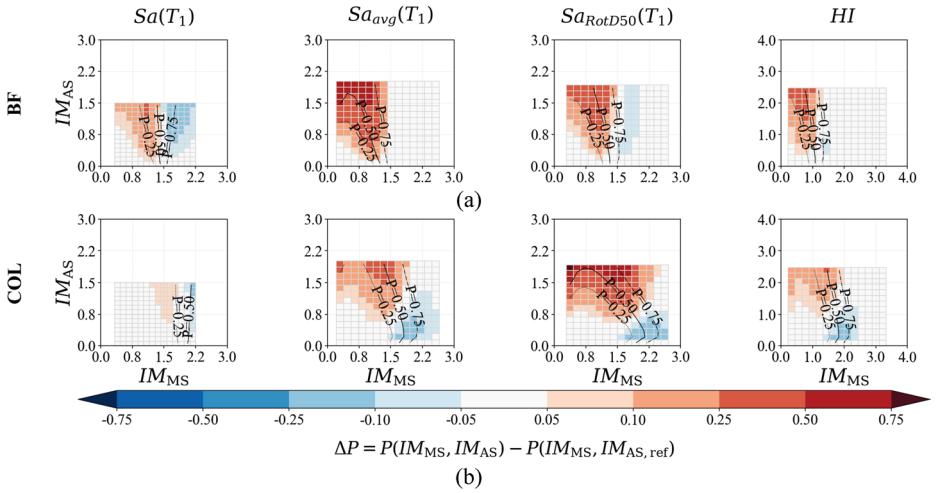

Fig. C.1 Heatmaps of the aftershock-induced exceedance-probability increment, (ΔP), for the four intensity measures $Sa(T_1)$, $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, and HI. Results are shown for: (**a**) BF at DS2 (δ_{max} = 0.152 m); and (**b**) COL at DS2 (DI = 0.40)

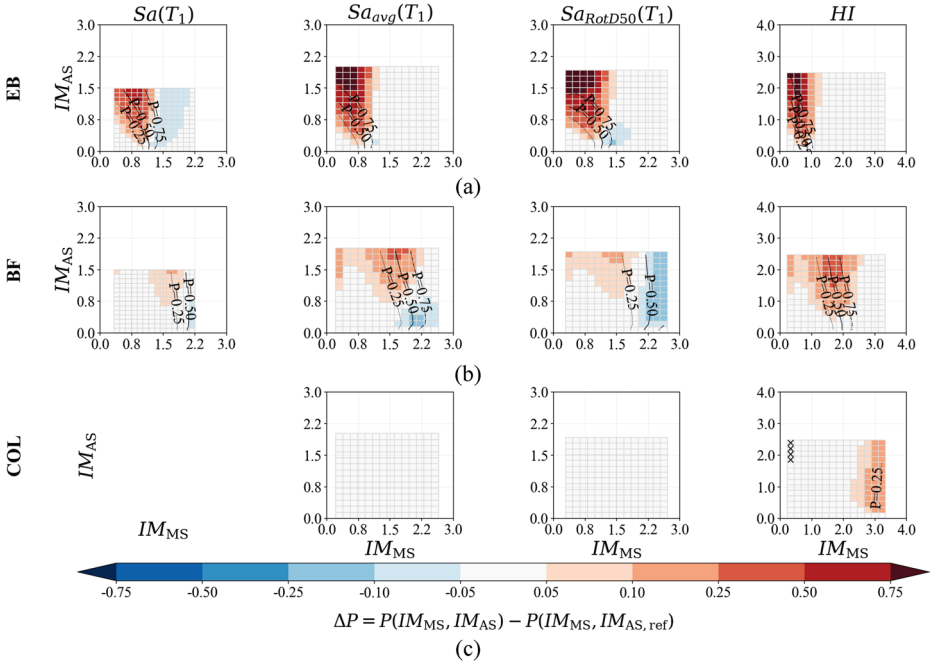

Fig. C.2 Heatmaps of the aftershock-induced exceedance-probability increment, (ΔP), for the four intensity measures $Sa(T_1)$, $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, and HI. Results are shown for: (**a**) EB at DS3 (δ_{max} = 0.136 m); (**b**) BF at DS3 (δ_{max} = 0.187 m); and (**c**) COL at DS3 (DI = 0.85)

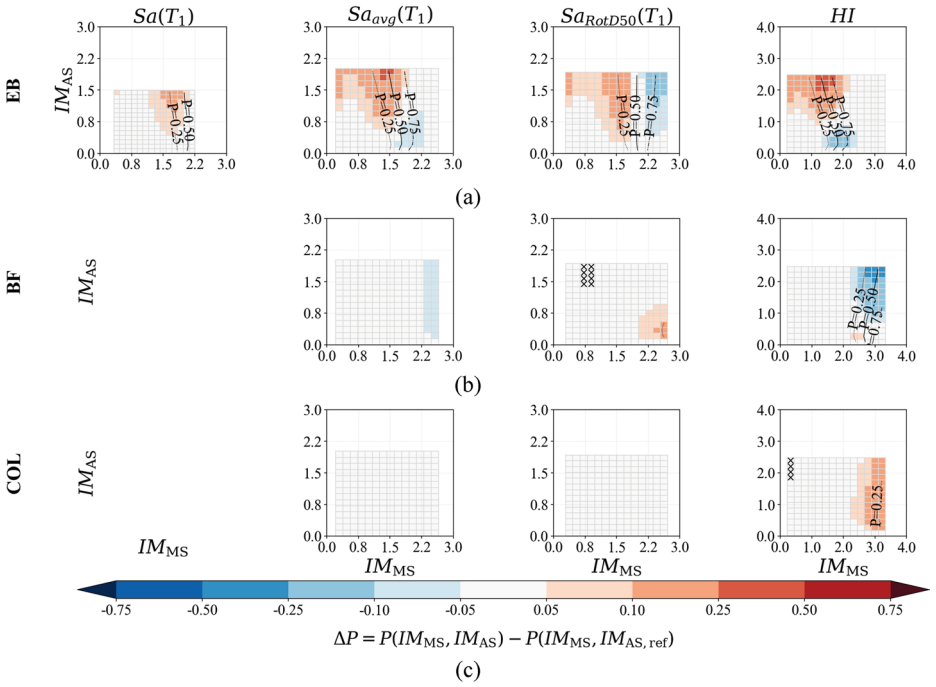

Fig. C.3 Heatmaps of the aftershock-induced exceedance-probability increment, (ΔP), for the four intensity measures $Sa(T_1)$, $Sa_{avg}(T_1)$, $Sa_{RotD50}(T_1)$, and HI. Results are shown for: (**a**) EB at DS4 (δ_{max} = 0.187 m); (**b**) BF at DS4 (δ_{max} = 0.278 m); and (**c**) COL at DS4 (DI = 1.00)

Table C.1 summarizes the mean and maximum ΔP values for all valid component–DS–IM combinations, providing a quantitative overview of the AS-induced increase in exceedance probability. Figure 31 and Table C.2 complement this probability-increment analysis through a contour-based interpretation of the AS effect. Table C.2 reports the controlling IM, the baseline MS intensity ($IM^{c}_{\mathrm{MS},p,\,base}$), the ($IM^{c}_{\mathrm{MS},p}$, $IM^{c}_{\mathrm{AS},p}$) pair at which $R_{IM_{\mathrm{MS}},p}$ reaches its maximum, the corresponding maximum reduction, and the remaining MS intensity percentage. Together, these supplementary results document the robustness of the AS-induced probability amplification across bridge components, DS, and candidate IMs.

Table C.1 Summary of mean and maximum aftershock-induced exceedance-probability increments, ΔP, for all valid component–damage-state–IM combinations

Component	DS	IM	Mean ΔP [pp]	Max. ΔP [pp]	IM_{MS} at max. ΔP	IM_{AS} at max. ΔP
EB	DS2	$Sa(T_1)$	12.403	82.916	0.672	1.443
		$Sa_{avg}(T_1)$	17.908	98.675	0.294	1.942
		$Sa_{RotD50}(T_1)$	11.892	93.681	0.372	1.855
		HI	11.304	95.195	0.331	2.393
	DS3	$Sa(T_1)$	7.959	59.793	0.81	1.443
		$Sa_{avg}(T_1)$	13.212	88.748	0.294	1.942
		$Sa_{RotD50}(T_1)$	15.716	92.907	0.372	1.855
		HI	9.945	84.267	0.553	2.393
	DS4	$Sa(T_1)$	3.284	13.109	1.772	1.443
		$Sa_{avg}(T_1)$	4.217	28.428	1.343	1.942
		$Sa_{RotD50}(T_1)$	2.668	16.779	0.372	1.855
		HI	3.988	35.053	1.441	2.393
BF	DS1	$Sa(T_1)$	2.583	38.976	0.81	1.443
		$Sa_{avg}(T_1)$	8.232	73.966	0.469	1.942
		$Sa_{RotD50}(T_1)$	13.721	82.954	0.718	1.855
		HI	9.024	82.829	0.553	2.393
	DS2	$Sa(T_1)$	2.537	26.929	1.085	1.443
		$Sa_{avg}(T_1)$	12.939	70.214	0.294	1.942
		$Sa_{RotD50}(T_1)$	6.702	43.902	0.891	1.855
		HI	3.593	37.172	0.775	2.393
	DS3	$Sa(T_1)$	2.177	10.639	1.635	1.443
		$Sa_{avg}(T_1)$	4.569	30.184	1.693	1.942
		$Sa_{RotD50}(T_1)$	0.127	11.533	1.583	1.855
		HI	7.925	37.125	1.663	2.393
	DS4	$Sa_{avg}(T_1)$	-1.443	2.106	0.294	1.942
		$Sa_{RotD50}(T_1)$	1.861	12.801	2.622	0.346
		HI	-3.256	5.451	2.552	0.268
COL	DS1	$Sa(T_1)$	6.338	40.819	1.085	1.443
		$Sa_{avg}(T_1)$	19.069	87.227	0.818	1.942
		$Sa_{RotD50}(T_1)$	13.506	78.762	0.372	1.855
		HI	12.183	85.444	0.553	2.393
	DS2	$Sa(T_1)$	1.432	7.225	1.635	1.443
		$Sa_{avg}(T_1)$	5.199	36.726	1.343	1.942
		$Sa_{RotD50}(T_1)$	14.567	76.344	0.372	1.855
		HI	2.18	25.002	1.441	2.393
	DS3	$Sa_{avg}(T_1)$	0.575	1.61	2.567	1.942
		$Sa_{RotD50}(T_1)$	0.62	1.9	2.622	1.855
		HI	4.144	23.116	3.218	0.799
	DS4	$Sa_{avg}(T_1)$	0.575	1.61	2.567	1.942
		$Sa4535_{RotD50}(T_1)$	0.62	1.9	2.622	1.855
		HI	4.144	23.116	3.218	0.799

Note: Values are expressed in percentage points (pp), not as relative percentages

For COL, identical ΔP values are reported for DS3 and DS4 because the exceedance sets obtained from the continuous column damage index used are identical for both damage states thresholds. In the analyzed NLTHA dataset, all records satisfying DI ≥ 0.85 also satisfy DI ≥ 1.00, and no records exist in the interval 0.85 ≤ DI < 1.00

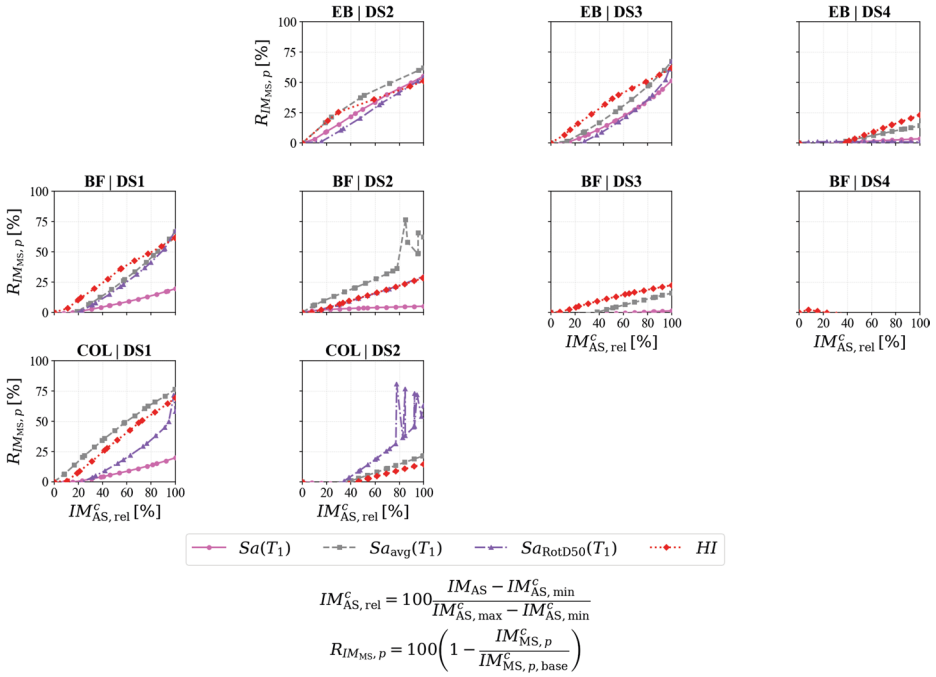

Fig. C.4 Contour-based reduction in the mainshock intensity measure required to maintain a fixed exceedance probability level, (P=0.50)

Table C.2 Summary of the maximum contour-based AS effect for fixed exceedance-probability levels

p	Component	DS	Controlling IM	IM units	$IM^c_{MS,p,base}$	$IM^c_{MS,p}$ at max R	$IM^c_{AS,p}$ at max R	Max $R_{IM_{MS},p}$ [%]	Remaining IM_{MS} [%]
0.5	EB	DS2	$Sa_{avg}(T_1)$	g	0.77	0.29	0.67	61.9	38.1
		DS3	$Sa_{avg}(T_1)$	g	0.90	0.29	1.15	67.4	32.6
		DS4	HI	m	1.69	1.30	2.39	22.9	77.1
	BF	DS1	$Sa_{RotD50}(T_1)$	g	1.13	0.37	1.27	67.2	32.8
		DS2	$Sa_{avg}(T_1)$	g	1.24	0.29	1.48	76.3	23.7
		DS3	HI	m	1.97	1.53	2.39	22.4	77.6
		DS4	HI	m	2.70	2.64	0.27	2	98
	COL	DS1	$Sa_{avg}(T_1)$	g	1.24	0.29	1.80	76.4	23.6
		DS2	$Sa_{RotD50}(T_1)$	g	1.94	0.37	1.46	80.9	19.1

Author contributions Brian Cagua: Formal analysis, Software, Methodology, Data curation, Visualization, Writing–original draft; Gerard J. O'Reilly: Conceptualization, Methodology, Supervision, Writing–review and editing; Jawad Fayaz: Software, Conceptualization, Methodology, Data curation, Writing–review and editing; Rodrigo Astroza: Conceptualization, Methodology, Supervision, Funding acquisition, Project administration, Resources, Writing–review and editing.

Funding This research was funded by the Chilean National Agency for Research and Development (ANID) through the grants FONDECYT 1240503 and Beca de Doctorado Nacional 21220089.

Data availability Data available on request from the authors.

Declarations

Conflict of interest All authors declare that they have no conflict of interest.

References

Abrahamson N, Gülerce Z (2020) Regionalized Ground-Motion Models for Subduction Earthquakes Based on the NGA-SUB Database. Pacific Earthquake Engineering Research Center, Berkeley, CA

Afsar Dizaj E, Salami MR, Kashani MM (2022) Seismic vulnerability assessment of ageing reinforced concrete structures under real mainshock-aftershock ground motions. Struct Infrastruct Eng 18:1674–1690. https://doi.org/10.1080/15732479.2021.1919148

Aoi S, Kunugi T, Fujiwara H (2004) Strong-motion seismograph network operated by NIED: K-NET and KiK-net. J Japan Association Earthq Eng 4:65–74. https://doi.org/10.5610/jaee.4.3_65

Araya R, Saragoni GR (1984) Earthquake accelerogram destructiveness potential factor. In: Proceedings of the 8th World Conference on Earthquake Engineering, San Francisco, USA, pp 835–842

Arias A (1970) A measure of earthquake intensity. Seismic Design for Nuclear Power Plants. MIT Press, Cambridge, MA, pp 438–483

Baker JW, Cornell C (2006) Spectral shape, epsilon and record selection. Earthq Eng Struct Dynamics 35:1077–1095. https://doi.org/10.1002/eqe.571

Barbosa AR, Ribeiro FLA, Neves LAC (2017) Influence of earthquake ground-motion duration on damage estimation: application to steel moment resisting frames. Earthq Eng Struct Dynamics 46:27–49. https://doi.org/10.1002/eqe.2769

Boore DM, Watson-Lamprey J, Abrahamson NA (2006) Orientation-Independent Measures of Ground Motion. Bull Seismol Soc Am 96:1502–1511. https://doi.org/10.1785/0120050209

Cagua B, Fayaz J, Astroza R (2026) Seismic fragility analysis considering mainshock-aftershock sequences: application to Chilean highway bridges. Eng Struct 360:122788. https://doi.org/10.1016/j.engstruct.2026.122788

Candia G, Macedo J, Jaimes MA, Magna-Verdugo C (2019) A New State-of-the-Art Platform for Probabilistic and Deterministic Seismic Hazard Assessment. Seismol Res Lett 90:2262–2275. https://doi.org/10.1785/0220190025

Castro S, Benavente R, Crempien JGF et al (2022) A Consistently Processed Strong-Motion Database for Chilean Earthquakes. Seismol Res Lett 93:2700–2718. https://doi.org/10.1785/0220200336

Chandramohan R, Baker JW, Deierlein GG (2016) Quantifying the Influence of Ground Motion Duration on Structural Collapse Capacity Using Spectrally Equivalent Records. Earthq Spectra 32:927–950. https://doi.org/10.1193/122813eqs298mr2

Chen X, Xiang N, Guan Z, Li J (2022) Seismic vulnerability assessment of tall pier bridges under mainshock-aftershock-like earthquake sequences using vector-valued intensity measure. Eng Struct 253:113732. https://doi.org/10.1016/j.engstruct.2021.113732

Chen L, Chen L, Zhou J (2024) Seismic fragility analysis of girder bridges under mainshock-aftershock sequences based on input-output hidden Markov model. Earthq Eng Struct Dynamics 53:3469–3488. https://doi.org/10.1002/eqe.4182

Cornell CA (1968) Engineering seismic risk analysis. Bull Seismol Soc Am 58:1583–1606

Cosenza E, Manfredi G, Ramasco R (1993) The use of damage functionals in earthquake engineering: A comparison between different methods. Earthq Eng Struct Dynamics 22:855–868. https://doi.org/10.1002/eqe.4290221003

Dávalos H, Miranda E (2020) Evaluation of FIV3 as an Intensity Measure for Collapse Estimation of Moment-Resisting Frame Buildings. J Struct Eng 146:04020204. https://doi.org/10.1061/(ASCE)ST.1943-541X.0002781

Davis SD, Frohlich C (1991) Single-link cluster analysis, synthetic earthquake catalogues, and aftershock identification. Geophys J Int 104:289–306. https://doi.org/10.1111/j.1365-246X.1991.tb02512.x

Di Trapani F, Malavisi M (2019) Seismic fragility assessment of infilled frames subject to mainshock/aftershock sequences using a double incremental dynamic analysis approach. Bull Earthq Eng 17:211–235. https://doi.org/10.1007/s10518-018-0445-2

Du A, Padgett JE, Shafieezadeh A (2019) A posteriori optimal intensity measures for probabilistic seismic demand modeling. Bull Earthq Eng 17:681–706. https://doi.org/10.1007/s10518-018-0484-8

Duncan JM, Mokwa RL (2001) Passive Earth Pressures: Theories and Tests. J Geotech GeoEnviron Eng 127:248–257. https://doi.org/10.1061/(ASCE)1090-0241(2001)127

Eads L, Miranda E, Lignos DG (2015) Average spectral acceleration as an intensity measure for collapse risk assessment. Earthq Eng Struct Dynamics 44:2057–2073. https://doi.org/10.1002/eqe.2575
Ebrahimian H, Jalayer F, Asprone D et al (2014) A performance-based framework for adaptive seismic aftershock risk assessment. Earthq Eng Struct Dynamics 43:2179–2197. https://doi.org/10.1002/eqe.2444
Fayaz J (2023) Select ground motions using weighted multi-IM objective. In: GitHub. https://github.com/jfayaz/Select_GMs_using_Weighted_Multi-IM_Objective. Accessed 13 Sept 2024
Fayaz J, Galasso C (2022) A generalized ground-motion model for consistent mainshock–aftershock intensity measures using successive recurrent neural networks. Bull Earthq Eng 20:6467–6486. https://doi.org/10.1007/s10518-022-01432-w
Fayaz J, Medalla M, Zareian F (2020) Sensitivity of the response of Box-Girder Seat-type bridges to the duration of ground motions arising from crustal and subduction earthquakes. Eng Struct 219:110845. https://doi.org/10.1016/j.engstruct.2020.110845
Fayaz J, Medalla M, Torres-Rodas P, Galasso C (2023) A recurrent-neural-network-based generalized ground-motion model for the Chilean subduction seismic environment. Struct Saf 100:102282. https://doi.org/10.1016/j.strusafe.2022.102282
Fragiacomo M, Amadio C, Macorini L (2004) Seismic response of steel frames under repeated earthquake ground motions. Eng Struct 26:2021–2035. https://doi.org/10.1016/j.engstruct.2004.08.005
Franco IEM, Smerzini C, Rosti A et al (2025) Seismic fragility curves with unconventional ground motion intensity measures from physics-based simulations. Bull Earthq Eng 23:1885–1915. https://doi.org/10.1007/s10518-025-02104-1
García de Quevedo P, Kohrangi M, Cattari S et al (2025) Considering cumulative damage in URM buildings for clustered seismicity risk assessment. Earthq Eng Struct Dynamics 54:1172–1190. https://doi.org/10.1002/eqe.4304
Gehl P, Seyedi DM, Douglas J (2013) Vector-valued fragility functions for seismic risk evaluation. Bull Earthq Eng 11:365–384. https://doi.org/10.1007/s10518-012-9402-7
Gidaris I, Padgett JE, Misra S (2022) Probabilistic fragility and resilience assessment and sensitivity analysis of bridges incorporating aftershock effects. Sustainable Resilient Infrastructure 7:17–39. https://doi.org/10.1080/23789689.2019.1708174
Goda K (2012) Nonlinear Response Potential of Mainshock–Aftershock Sequences from Japanese Earthquakes. Bull Seismol Soc Am 102:2139–2156. https://doi.org/10.1785/0120110329
Goel RK, Chopra AK (2008) Role of Shear Keys in Seismic Behavior of Bridges Crossing Fault-Rupture Zones. J Bridge Engineering 13:398–408. https://doi.org/10.1061/(ASCE)1084-0702(2008)13:4(398)
Guajardo B, Pinto F, Astroza R (2024) Effects of soil spatial variability on the seismic response of multi-span simply-supported highway bridges. Bull Earthq Eng 22:2643–2675. https://doi.org/10.1007/s10518-024-01872-6
Guo X, Zhang Z, Chen Z (2020) Mainshock-Integrated Aftershock Vulnerability Assessment of Bridge Structures. Appl Sci 10:6843. https://doi.org/10.3390/app10196843
Gutenberg B, Richter CF (1942) Earthquake magnitude, intensity, energy, and acceleration. Bull Seismol Soc Am 32:163–191. https://doi.org/10.1785/BSSA0320030163
Gutenberg B, Richter CF (1944) Frequency of earthquakes in California. Bull Seismol Soc Am 34:185–188. https://doi.org/10.1785/BSSA0340040185
Hancock J, Bommer JJ (2006) A State-of-Knowledge Review of the Influence of Strong-Motion Duration on Structural Damage. Earthq Spectra 22:827–845. https://doi.org/10.1193/1.2220576
Harati M, van de Lindt JW (2024) Mainshock-aftershock building fragility methodology for community resilience modeling. Structures 70:107742. https://doi.org/10.1016/j.istruc.2024.107742
Hatzigeorgiou GD, Beskos DE (2009) Inelastic displacement ratios for SDOF structures subjected to repeated earthquakes. Eng Struct 31:2744–2755. https://doi.org/10.1016/j.engstruct.2009.07.002
Hayes GP, Moore GL, Portner DE et al (2018) Slab2, a comprehensive subduction zone geometry model. Science 362:58–61. https://doi.org/10.1126/science.aat4723
Housner G (1952) Spectrum intensities of strong-motion earthquakes. In: Proceedings of the Symposium on Earthquake and Blast Effects on Structures. Earthquake Engineering Research Institute, pp 20–36
Idini B, Rojas F, Ruiz S, Pastén C (2017) Ground motion prediction equations for the Chilean subduction zone. Bull Earthq Eng 15:1853–1880. https://doi.org/10.1007/s10518-016-0050-1
Jalayer F, Ebrahimian H (2017) Seismic risk assessment considering cumulative damage due to aftershocks. Earthq Eng Struct Dynamics 46:369–389. https://doi.org/10.1002/eqe.2792
Jiang L, Zhong J, He M, Yuan W (2019) Optimal Seismic Intensity Measure Selection for Isolated Bridges under Pulse-Like Ground Motions. Adv Civil Eng 2019:3858457. https://doi.org/10.1155/2019/3858457
Kagan YY (2002) Aftershock Zone Scaling. Bull Seismol Soc Am 92:641–655. https://doi.org/10.1785/0120010172
Kawashima K, Buckle I (2013) Structural Performance of Bridges in the Tohoku-Oki Earthquake. Earthq Spectra 29:315–338. https://doi.org/10.1193/1.4000129

Kawashima K, Unjoh S, Hoshikuma J-I, Kosa K (2011) Damage of Bridges due to the 2010 Maule, Chile, Earthquake. J Earthquake Eng 15:1036–1068. https://doi.org/10.1080/13632469.2011.575531
Kourehpaz P, Molina Hutt C, Lallemant D (2023) Toward multivariate fragility functions for seismic damage and loss estimation of high-rise buildings. Earthq Eng Struct Dynamics 52:4164–4182. https://doi.org/10.1002/eqe.3993
Kuehn NM, Abrahamson NA (2020) Spatial correlations of ground motion for non-ergodic seismic hazard analysis. Earthq Eng Struct Dynamics 49:4–23. https://doi.org/10.1002/eqe.3221
Lee K, Foutch D (2004) Performance Evaluation of Damaged Steel Frame Buildings Subjected to Seismic Loads. J Struct Eng 130:588–599. https://doi.org/10.1061/(ASCE)0733-9445(2004)130:4(588)
Lei X, Sun L, Xia Y (2021) Seismic fragility assessment and maintenance management on regional bridges using bayesian multi-parameter estimation. Bull Earthq Eng 19:6693–6717. https://doi.org/10.1007/s10518-021-01072-6
Li Q, Ellingwood BR (2007) Performance evaluation and damage assessment of steel frame buildings under main shock–aftershock earthquake sequences. Earthq Eng Struct Dynamics 36:405–427. https://doi.org/10.1002/eqe.667
Li Yue S, Van De Ruiqiang W (2014) Collapse Fragility of Steel Structures Subjected to Earthquake Mainshock-Aftershock Sequences. J Struct Eng 140:04014095. https://doi.org/10.1061/(ASCE)ST.1943-541X.0001019
Liu J, Tian L, Meng X, Yang M (2022) Seismic fragility assessment of a transmission tower considering mainshock-aftershock sequences. J Constr Steel Res 194:107344. https://doi.org/10.1016/j.jcsr.2022.107344
Liu J, Meng X, Tian L et al (2024a) Multi-metric evaluation of the optimal intensity measure for mainshock-aftershock fragility analysis of transmission towers. Eng Struct 308:118020. https://doi.org/10.1016/j.engstruct.2024.118020
Liu J, Tian L, Yang M, Meng X (2024b) Probabilistic framework for seismic resilience assessment of transmission tower-line systems subjected to mainshock-aftershock sequences. Reliab Eng Syst Saf 242:109755. https://doi.org/10.1016/j.ress.2023.109755
Liu J, Tian L, Zhang R et al (2025) Performance-based failure risk evaluation of transmission tower-line systems subjected to sequential earthquakes. J Constr Steel Res 234:109730. https://doi.org/10.1016/j.jcsr.2025.109730
Luco N, Cornell CA (2007) Structure-Specific Scalar Intensity Measures for Near-Source and Ordinary Earthquake Ground Motions. Earthq Spectra 23:357–392. https://doi.org/10.1193/1.2723158
Luco N, Bazzurro P, Cornell CA (2004) Dynamic versus static computation of the residual capacity of a mainshock-damaged building to withstand an aftershock. In: Proceedings of the 13th World Conference on Earthquake Engineering. Vancouver, Canada
Luco N, Gerstenberger MC, Uma S et al (2011) A methodology for post-mainshock probabilistic assessment of building collapse risk. In: Proceedings of the 9th Pacific Conference on Earthquake Engineering, Auckland, New Zealand
Mackie K, Stojadinović B (2001) Probabilistic Seismic Demand Model for California Highway Bridges. J Bridge Engineering 6:468–481. https://doi.org/10.1061/(ASCE)1084-0702(2001)6:6(468)
Martínez Grosser A (2016) Efecto de las barras sísmicas en el comportamiento sísmico transversal de puentes de hormigón armado (In Spanish). MSc Thesis. Pontificia Universidad Católica de Chile
McKenna F, Fenves GL, Scott MH (2000) Open system for earthquake engineering simulation. University of California, Berkeley, CA
Montalva GA, Bastías N, Rodriguez-Marek A (2017) Ground-Motion Prediction Equation for the Chilean Subduction Zone. Bull Seismol Soc Am 107:901–911. https://doi.org/10.1785/0120160221
Newmark NM, Hall WJ (1973) Procedures and criteria for earthquake-resistant design. Selected Papers By Nathan M. Civil Engineering Classics. ASCE, Newmark, pp 829–872
Newmark NM, Hall WJ (1982) Earthquake spectra and design. Engineering monographs on earthquake criteria
Nicolis O, Chiodi M, Adelfio G (2015) Windowed ETAS models with application to the Chilean seismic catalogs. Spat Stat 14:151–165. https://doi.org/10.1016/j.spasta.2015.05.006
O'Reilly GJ (2021a) Limitations of Sa(T1) as an intensity measure when assessing non-ductile infilled RC frame structures. Bull Earthq Eng 19:2389–2417. https://doi.org/10.1007/s10518-021-01071-7
O'Reilly GJ (2021b) Seismic intensity measures for risk assessment of bridges. Bull Earthq Eng 19:3671–3699. https://doi.org/10.1007/s10518-021-01114-z
Ogata Y (1998) Space-Time Point-Process Models for Earthquake Occurrences. Ann Inst Stat Math 50:379–402. https://doi.org/10.1023/A:1003403601725
Ogata Y, Zhuang J (2006) Space–time ETAS models and an improved extension. Tectonophysics 413:13–23. https://doi.org/10.1016/j.tecto.2005.10.016
Omori F (1895) On the After-shocks of Earthquakes. J Coll Sci Imperial Univ Japan 7:111–200. https://doi.org/10.15083/00037562

Orlacchio M, Baltzopoulos G, Iervolino I (2024) Simplified state-dependent seismic fragility assessment. Earthq Eng Struct Dynamics 53:2099–2121. https://doi.org/10.1002/eqe.4105
Otárola K, Fayaz J, Galasso C (2022) Fragility and vulnerability analysis of deteriorating ordinary bridges using simulated ground-motion sequences. Earthq Eng Struct Dynamics 51:3215–3240. https://doi.org/10.1002/eqe.3720
Padgett JE, Nielson BG, DesRoches R (2008) Selection of optimal intensity measures in probabilistic seismic demand models of highway bridge portfolios. Earthq Eng Struct Dynamics 37:711–725. https://doi.org/10.1002/eqe.782
Park Y, Ang AH-S (1985) Mechanistic Seismic Damage Model for Reinforced Concrete. J Struct Eng 111:722–739. https://doi.org/10.1061/(ASCE)0733-9445(1985)111
Parker G, Stewart J, Boore D et al (2021) NGA-subduction global ground motion models with regional adjustment factors. Earthq Spectra 38:456–493. https://doi.org/10.1177/87552930211034889
Pasten C, Astroza R, Bazaez R et al (2021) Guía para el Análisis Sísmico No Lineal de Puentes Chilenos. Santiago, Chile
PEER Ground Motion Database - PEER Center https://ngawest2.berkeley.edu/. Accessed 12 Sept 2024
Pinto F, Astroza R, Bazáez R et al (2024) Probabilistic seismic assessment of multispan RC highway bridges considering soil-structure interaction and chloride-induced corrosion. Eng Struct 301:117257. https://doi.org/10.1016/j.engstruct.2023.117257
Poulos A, Monsalve M, Zamora N, de la Llera JC (2018) An Updated Recurrence Model for Chilean Subduction Seismicity and Statistical Validation of Its Poisson Nature. Bull Seismol Soc Am 109:66–74. https://doi.org/10.1785/0120170160
Raghunandan M, Liel AB (2013) Effect of ground motion duration on earthquake-induced structural collapse. Struct Saf 41:119–133. https://doi.org/10.1016/j.strusafe.2012.12.002
Reed JW, Kassawara RP (1990) A criterion for determining exceedance of the operating basis earthquake. Nucl Eng Des 123:387–396. https://doi.org/10.1016/0029-5493(90)90259-Z
Rubilar Moya F (2016) Modelo no lineal para predecir la respuesta sísmica de pasos superiores (In Spanish). MSc Thesis. Pontificia Universidad Católica de Chile
Ruiz-García J (2012) Mainshock-Aftershock Ground Motion Features and Their Influence in Building's Seismic Response. J Earthquake Eng 16:719–737. https://doi.org/10.1080/13632469.2012.663154
Ruiz-García J, Negrete-Manriquez JC (2011) Evaluation of drift demands in existing steel frames under as-recorded far-field and near-fault mainshock–aftershock seismic sequences. Eng Struct 33:621–634. https://doi.org/10.1016/j.engstruct.2010.11.021
Ryu H, Luco N, Uma S, Liel A (2011) Developing fragilities for mainshock-damaged structures through incremental dynamic analysis. In: Proceedings of the 9th Pacific Conference on Earthquake Engineering, Auckland, New Zealand
Saichev A, Sornette D (2005) Distribution of the largest aftershocks in branching models of triggered seismicity: Theory of the universal Båth law. Phys Rev E 71:056127. https://doi.org/10.1103/PhysRevE.71.056127
Shirato M, Odawara Y (2024) Field observations on the damage to road bridges after the 2023 Turkey Earthquake. Earthq Spectra 40:997–1018. https://doi.org/10.1177/87552930241234271
Shokrabadi M, Burton HV (2018) Risk-based assessment of aftershock and mainshock-aftershock seismic performance of reinforced concrete frames. Struct Saf 73:64–74. https://doi.org/10.1016/j.strusafe.2018.03.003
Teran-Gilmore A, Jirsa JO (2005) A Damage Model for Practical Seismic Design that Accounts for Low Cycle Fatigue. Earthq Spectra 21:803–832. https://doi.org/10.1193/1.1979500
Trifunac MD, Brady AG (1975) A study on the duration of strong earthquake ground motion. Bull Seismol Soc Am 65:581–626. https://doi.org/10.1785/BSSA0650030581
Utsu T, Ogata Y, Matsu'ura SR (1995) The Centenary of the Omori Formula for a Decay Law of Aftershock Activity. J Phys Earth 43:1–33. https://doi.org/10.4294/jpe1952.43.1
Vamvatsikos D, Cornell CA (2002) Incremental dynamic analysis. Earthq Eng Struct Dynamics 31:491–514. https://doi.org/10.1002/eqe.141
Wang X, Shafieezadeh A, Ye A (2018) Optimal intensity measures for probabilistic seismic demand modeling of extended pile-shaft-supported bridges in liquefied and laterally spreading ground. Bull Earthq Eng 16:229–257. https://doi.org/10.1007/s10518-017-0199-2
Wang T, Han Q, Wen J, Wang L (2022) Analysis of the Effect of Mainshock-Aftershock Sequences on the Fragility of RC Bridge Columns. Buildings 12:1681. https://doi.org/10.3390/buildings12101681
Wen W, Zhai C, Ji D et al (2017) Framework for the vulnerability assessment of structure under mainshock-aftershock sequences. Soil Dyn Earthq Eng 101:41–52. https://doi.org/10.1016/j.soildyn.2017.07.002
Wiens DA, McGuire JJ (1995) The 1994 Bolivia and Tonga events: Fundamentally different types of deep earthquakes? Geophys Res Lett 22:2245–2248. https://doi.org/10.1029/95GL01598

Williams MS, Sexsmith RG (1995) Seismic Damage Indices for Concrete Structures: A State-of-the-Art. Rev Earthq spectra 11:319–349. https://doi.org/10.1193/1.1585817

Yeo GL, Cornell CA (2005) Stochastic characterization and decision bases under time-dependent aftershock risk in performance-based earthquake engineering. Pacific Earthquake Engineering Research Center, University of California, Berkeley, CA

Yeo GL, Cornell CA (2009) A probabilistic framework for quantification of aftershock ground-motion hazard in California: Methodology and parametric study. Earthq Eng Struct Dynamics 38:45–60. https://doi.org/10.1002/eqe.840

Yu X, Zhou Z, Du W, Lu D (2021) Development of fragility surfaces for reinforced concrete buildings under mainshock-aftershock sequences. Earthq Eng Struct Dynamics 50:3981–4000. https://doi.org/10.1002/eqe.3542

Yu X-H, Zhou Z, Lu D-G, Ji K (2024) A Practical Approach of Probabilistic Seismic Hazard Analysis for Vector IMs Regarding Mainshock with Potentially Largest Aftershock. J Earthquake Eng 28:637–658. https://doi.org/10.1080/13632469.2023.2217723

Zhang L, Werner MJ, Goda K (2020) Variability of ETAS Parameters in Global Subduction Zones and Applications to Mainshock–Aftershock Hazard Assessment. Bull Seismol Soc Am 110:191–212. https://doi.org/10.1785/0120190121

Zhong J, Pang Y, Jeon J-S et al (2016) Seismic fragility assessment of long-span cable-stayed bridges in China. Adv Struct Eng 19:1797–1812. https://doi.org/10.1177/1369433216649380

Zhou Z, Yu X, Lu D (2020) Identifying Optimal Intensity Measures for Predicting Damage Potential of Mainshock–Aftershock Sequences. Appl Sci 10:6795. https://doi.org/10.3390/app10196795

Zhou Z, Xu H, Gardoni P et al (2021) Probabilistic demand models and fragilities for reinforced concrete frame structures subject to mainshock-aftershock sequences. Eng Struct 245:112904. https://doi.org/10.1016/j.engstruct.2021.112904

Zhou W, Xiong L, Jiang L et al (2024) Optimizing seismic fragility assessments for high-speed railway track bridges: a novel multi-parameter methodology and intensity measure selection. Mech Based Des Struct Mach 52:7159–7179. https://doi.org/10.1080/15397734.2023.2301378

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Brian Cagua[1] · Gerard J. O'Reilly[2] · Jawad Fayaz[3] · Rodrigo Astroza[1]

✉ Rodrigo Astroza
rastroza@miuandes.cl

[1] Facultad de Ingeniería y Ciencias Aplicadas, Universidad de los Andes, Santiago, Chile

[2] Centre for Training and Research on Reduction of Seismic Risk (ROSE Centre), Scuola Universitaria Superiore IUSS Pavia, Pavia, Italy

[3] Department of Computer Science, University of Exeter, Exeter EX4 4RN, UK